

3 TIMED TESTS · 50 QUESTIONS EACH

Neurology

Practice tests of USMLE-style questions drawn from the website quizzes, with answers and explanations.

For educational use only. This is a study aid, not a clinical reference.

Instructions

60 minutes
for each test

50 questions
in each test

About 72 seconds
for each question

Every question comes from the website. These tests are built from the quizzes on the Clinical Conditions website, 150 of its 437 questions, so if you have done the website quizzes you will meet questions you have seen before. For the truest measure, take a test before using the website quiz for these topics, and treat a remembered answer as revision rather than a sign of mastery.

TAKING A TEST

- Take each test in one sitting, with a timer set to 60 minutes. Do not pause the timer.
- Each question has five options and one best answer. Choose the single best option, even if more than one seems reasonable.
- Fill in the answer sheet after the last question of each test, or use a separate piece of paper.
- Do not use notes, the website, or other references. The abbreviation list on the next page is allowed.
- There is no penalty for guessing, so answer every question. An unanswered question counts as incorrect.
- When time ends, stop, even if you have not finished.

ABOUT THE TESTS

- No question appears in more than one test, so the three tests can be taken in any order.
- Every test has 15 physiology and 35 clinical questions, covering all 12 physiology topics and all 23 clinical topics, with a similar mix of Step 1 questions (mechanisms, microbiology, and pathology) and Step 2 questions (diagnosis, next step, and management).
- Questions are in random order, and the topic being tested is not shown until the answers.

SCORING AND REVIEW

- Your score is the number of correct answers. The answer key at the start of each answer section has the same layout as the answer sheet, so you can mark one against the other.
- Then read the explanations. Each one names the topic and step, so you can see what to revisit.
- Answers follow the course material. Where a guideline differs, the explanation says so under "Additional information".

CONTENTS

Test A	**Questions**	Test B	**Questions**	Test C	**Questions**
Test A	**Answer sheet**	Test B	**Answer sheet**	Test C	**Answer sheet**
Test A	**Answers**	Test B	**Answers**	Test C	**Answers**

Abbreviations

ALLOWED DURING A TEST

AComm Anterior communicating artery

ADC Apparent diffusion coefficient

AICA Anterior inferior cerebellar artery

ALS Amyotrophic lateral sclerosis

AMPA α-Amino-3-hydroxy-5-methyl-4-isoxazolepropionic acid

ANA Antinuclear antibody

ANP Atrial natriuretic peptide

APOE Apolipoprotein E

APP Amyloid precursor protein

AQP4 Aquaporin-4

AQP4-IgG Aquaporin-4 immunoglobulin G antibody

ARIA Amyloid-related imaging abnormalities

Aβ42 Amyloid beta 42, the 42-amino-acid form

B12 Vitamin B12 (cobalamin)

BMI Body mass index

BP Blood pressure

BPPV Benign paroxysmal positional vertigo

bvFTD Behavioral variant frontotemporal dementia

C9orf72 Chromosome 9 open reading frame 72 gene

CBD Corticobasal degeneration

CK Creatine kinase

CMV Cytomegalovirus

CN Cranial nerve

CN III Cranial nerve III (oculomotor)

CN VI Cranial nerve VI (abducens)

CN VII Cranial nerve VII (facial)

CN X Cranial nerve X (the vagus nerve)

COMT Catechol-O-methyltransferase

COPD Chronic obstructive pulmonary disease

CRAO Central retinal artery occlusion

CRPS Complex regional pain syndrome

CRVO Central retinal vein occlusion

CSF Cerebrospinal fluid

CT Computed tomography

CXR Chest x-ray

D1 Dopamine receptor 1

D2 Dopamine receptor 2

DAT Dopamine transporter

DBS Deep brain stimulation

DLB Dementia with Lewy bodies

DOAC Direct oral anticoagulant

DWI Diffusion-weighted imaging

ECG Electrocardiogram

EDNRB Endothelin receptor type B gene

EEG Electroencephalogram

ESR Erythrocyte sedimentation rate

FLAIR Fluid-attenuated inversion recovery

GCS Glasgow Coma Scale

GJB2 Gap junction beta 2 gene (connexin 26)

GPe Globus pallidus externus

GPi Globus pallidus internus

HINTS Head impulse, nystagmus, and test of skew

HLA-B*1502 Human leukocyte antigen B*15:02 allele

HR Heart rate

HTT Huntingtin gene

ICA Internal carotid artery

ICH Intracerebral hemorrhage

INO Internuclear ophthalmoplegia

IOP Intraocular pressure

IV Intravenous

IVIG Intravenous immunoglobulin

M1 Muscarinic acetylcholine receptor 1; Primary motor cortex

MAO-B Monoamine oxidase B

MCA Middle cerebral artery

MITF Melanocyte-inducing transcription factor gene

MLF Medial longitudinal fasciculus

MOG-IgG Myelin oligodendrocyte glycoprotein immunoglobulin G antibody

MOGAD MOG antibody-associated disease

MRA Magnetic resonance angiography

MRI Magnetic resonance imaging

MS Multiple sclerosis

NMDA N-methyl-D-aspartate

NMOSD Neuromyelitis optica spectrum disorder

NPH Normal pressure hydrocephalus

OCT Optical coherence tomography

PaCO2 Arterial partial pressure of carbon dioxide

PAX3 Paired box 3 gene

PComm Posterior communicating artery

PET Positron emission tomography

PFO Patent foramen ovale

PICA Posterior inferior cerebellar artery

POTS Postural orthostatic tachycardia syndrome

PSP Progressive supranuclear palsy

PSP-bvFTD Progressive supranuclear palsy, behavioral variant frontotemporal dementia

PSP-CBS Progressive supranuclear palsy, corticobasal syndrome

PSP-P Progressive supranuclear palsy, parkinsonism predominant

PSP-PNFA Progressive supranuclear palsy, progressive nonfluent aphasia

PSP-RS Progressive supranuclear palsy, Richardson syndrome

RAAS Renin-angiotensin-aldosterone system

RAPD Relative afferent pupillary defect

RTX Rituximab

SAH Subarachnoid hemorrhage

SNc Substantia nigra pars compacta

SNHL Sensorineural hearing loss

SNr Substantia nigra pars reticulata

SPECT Single-photon emission computed tomography

STN Subthalamic nucleus

SWI Susceptibility-weighted imaging

TPMT Thiopurine methyltransferase

V1 Ophthalmic division of the trigeminal nerve; Primary visual cortex

VA Ventral anterior nucleus of the thalamus

VEGF Vascular endothelial growth factor

VL Ventral lateral nucleus of the thalamus

Test A

50 questions | 60 minutes | Start the timer

1 A 63-year-old man has mild memory problems, and the diagnosis is uncertain after clinical assessment and MRI.

Which CSF pattern supports Alzheimer disease?

- (A) Increased Aβ42 with decreased tau and phosphorylated tau
- (B) Oligoclonal bands with raised white cells
- (C) Raised 14-3-3 protein only
- (D) Decreased Aβ42 with increased tau and phosphorylated tau
- (E) Normal Aβ42 and normal tau

2 A 33-year-old woman with MS notices that her leg weakness and blurred vision worsen after a hot shower and settle within an hour.

What is this phenomenon?

- (A) Lhermitte sign
- (B) Uhthoff phenomenon
- (C) An acute relapse
- (D) Charcot triad
- (E) Secondary progression

3 A 6-week-old infant has a white pupil in the right eye on a photograph. The red reflex is absent on that side.

Why is early treatment important?

- (A) To prevent retinal detachment
- (B) To prevent amblyopia
- (C) Because the lens will otherwise swell and close the angle
- (D) To prevent phacolytic glaucoma
- (E) Because congenital cataracts become malignant

4 A neurosurgeon traces the lateral corticospinal tract from M1 toward the brainstem.

Through which part of the internal capsule does it pass?

- (A) Retrolenticular part
- (B) External capsule
- (C) Genu
- (D) Posterior limb
- (E) Anterior limb

5 Morphine binds μ receptors on nociceptive terminals in the dorsal horn.

Which cellular effects reduce transmission?

- (A) Opening potassium channels and closing N-type calcium channels
- (B) Raising cAMP in the terminal through a Gs protein
- (C) Opening sodium channels and closing potassium channels in the terminal
- (D) Opening chloride channels like GABA-A
- (E) Blocking NMDA receptors

6 On day 5 after aneurysmal SAH, a 56-year-old woman's sodium falls to 127 mmol/L.

Which is the most appropriate approach?

- (A) Treat the cause and avoid fluid restriction
- (B) Give a loop diuretic
- (C) Strict fluid restriction to 800 mL a day until sodium recovers
- (D) Stop nimodipine
- (E) Hypotonic IV fluids

7 A 30-year-old man has orofacial chorea that worsens when he eats, tongue biting, and raised CK. His HTT repeat length is normal, and his blood film shows spiky red cells. His parents are unaffected.

What is the most likely diagnosis?

- (A) Hemiballismus from a stroke
- (B) Juvenile Huntington disease
- (C) Chorea-acanthocytosis
- (D) Spinocerebellar ataxia
- (E) Wilson disease

8 A 28-year-old woman has 3 days of blurred vision in her left eye and pain when she moves it. Colors look washed out, and she has a left RAPD. The optic disc looks normal.

Which investigation is most important next?

- (A) MRI of the brain and orbits with gadolinium
- (B) Electroretinogram
- (C) Fluorescein angiography of the retina and choroid
- (D) Gonioscopy
- (E) Temporal artery biopsy

9 A 60-year-old woman has lost the left half of the visual field in both eyes, without macular sparing.

Which lesion fits?

- (A) Right optic nerve
- (B) Right optic tract
- (C) Left Meyer loop
- (D) Left optic tract
- (E) Optic chiasm

10 A 62-year-old smoker has proximal leg weakness, dry mouth, and reduced reflexes that briefly improve after repeated contraction. CXR shows a hilar mass.

What is the target of the autoantibody?

- (A) Acetylcholinesterase
- (B) Voltage-gated sodium channels
- (C) Presynaptic voltage-gated calcium channels
- (D) Muscle-specific kinase
- (E) Postsynaptic nicotinic acetylcholine receptors

11 A 27-year-old woman has episodes that begin with a rising feeling in her stomach and intense déjà vu, followed by staring, chewing movements, and no memory of the event. Afterward she is confused for 20 minutes.

Where do these seizures most likely start?

- (A) Frontal lobe motor cortex
- (B) Occipital lobe
- (C) Cerebellum
- (D) Temporal lobe
- (E) Both hemispheres at once

12 A 66-year-old vegan woman stomps when she walks and is worse in the dark. She stands steadily with her eyes open but sways and nearly falls when she closes them.

Which pathway is damaged?

- (A) Dorsal columns
- (B) Cerebellar vermis
- (C) Corticospinal tracts
- (D) Basal ganglia
- (E) Right cerebellar hemisphere

13 A cardiac glycoside partially blocks the pump that maintains neuronal ion gradients.

What does this pump move with each cycle?

- (A) 2 sodium out and 3 potassium in
- (B) 1 calcium out and 3 sodium in
- (C) 1 sodium in and 1 chloride in
- (D) 3 potassium out and 2 sodium in
- (E) 3 sodium out and 2 potassium in

14 A woman thought to have NMOSD has CSF oligoclonal bands, short peripheral cord lesions, and periventricular lesions at right angles to the ventricles.

Which diagnosis do these findings favor?

- (A) MOGAD
- (B) Spinal cord infarction
- (C) Area postrema syndrome
- (D) AQP4-IgG-positive NMOSD
- (E) Multiple sclerosis

15 A 61-year-old woman speaks fluently but cannot name common objects and asks "what is a fork?" when told to pick one up.

Atrophy of which region is most likely?

- (A) Left anterior temporal lobe
- (B) Right anterior temporal lobe
- (C) Left inferior frontal lobe and insula
- (D) Occipital cortex
- (E) Bilateral hippocampi

16 A 78-year-old woman with progressive memory loss has a headache and left homonymous hemianopia. CT shows a right occipital lobar hemorrhage. MRI with SWI shows many lobar microbleeds. She has no history of hypertension.

What is the most likely cause?

(A) Cerebral venous sinus thrombosis
(B) Cerebral amyloid angiopathy
(C) Arteriovenous malformation
(D) Hypertension
(E) Mycotic aneurysm

17 In Parkinson disease, SNc dopamine neurons die.

What happens to GPi output?

(A) It increases, inhibiting the thalamus and causing bradykinesia
(B) It is unchanged
(C) It becomes excitatory
(D) It stops entirely, causing ballismus
(E) It decreases, releasing the VA and VL thalamus and causing chorea

18 After a stab wound to the right side of the back at T8, a man has weakness and loss of vibration and position sense in the right leg.

Where else is sensation lost?

(A) All sensation in both legs
(B) Pain and temperature in the left leg
(C) Pain and temperature in both arms
(D) Vibration in the left leg
(E) Pain and temperature in the right leg

19 A 60-year-old man has a small pontine stroke. Neither eye can look to the left, but convergence is preserved.

Which structure is damaged?

(A) Right MLF
(B) Left CN VI nerve
(C) Right frontal eye field
(D) Left CN III nucleus
(E) Left CN VI nucleus

20 A 41-year-old man woke 2 days ago with severe hearing loss in his left ear. Otoscopy is normal, and audiometry shows a 50 dB sensorineural loss across 4 consecutive frequencies.

Which is the most appropriate treatment?

(A) Oral amoxicillin
(B) Oral prednisone 1 mg/kg for 7 days, then tapered
(C) Microsuction
(D) A hearing aid fitted once his hearing stabilizes in 3 months
(E) Observation for 3 months

21 A physiology student rides a lift that accelerates upward.

Which structure senses this vertical linear acceleration?

(A) Horizontal canal
(B) Utricle
(C) Posterior canal
(D) Cochlea
(E) Saccule

22 A 59-year-old man with diabetes has burning pain in both feet that disturbs his sleep. He also has depression.

Which drug treats both, by strengthening descending inhibition?

(A) Paracetamol
(B) Ketamine
(C) Ibuprofen
(D) Duloxetine
(E) Morphine

23 A 64-year-old woman meets criteria for probable corticobasal syndrome, with asymmetric limb rigidity, dystonia, limb apraxia, and alien limb.

Which statement about her underlying pathology is correct?

(A) DAT SPECT confirms CBD pathology and excludes PSP
(B) It is always caused by α-synuclein in oligodendrocytes
(C) PSP, Alzheimer disease, or other pathologies can cause it
(D) It is always caused by CBD
(E) MRI shows the hot cross bun sign

24 A 67-year-old man arrives 50 minutes after sudden right face and arm weakness and difficulty speaking. His BP is 176/98 mm Hg.

Which of the following should be done first?

- (A) MRI with diffusion-weighted imaging
- (B) Glucose and non-contrast CT head
- (C) Aspirin 300 mg
- (D) Lower his BP to below 140/90
- (E) IV alteplase

25 A 70-year-old man has sudden left hemiplegia and neglect. His eyes are deviated to the right.

Where is the lesion?

- (A) Left hemisphere
- (B) Left pontine gaze center
- (C) Right hemisphere
- (D) Right medial longitudinal fasciculus
- (E) Dorsal midbrain

26 A 78-year-old woman admitted with a right MCA stroke is hungry and asks for breakfast. She has mild dysarthria.

What should happen before she eats or drinks?

- (A) Carotid duplex
- (B) Bubble echocardiogram
- (C) EEG
- (D) A swallow screen
- (E) Repeat CT head

27 An 84-year-old woman became confused over 2 days. Her attention wanders, she is worse at night, and she has a urinary tract infection.

What is the most likely cause of her confusion?

- (A) Delirium
- (B) Dementia with Lewy bodies
- (C) Depression
- (D) Alzheimer disease
- (E) Frontotemporal dementia

28 A 50-year-old woman has constant vertigo and vomiting for 1 day, with no other neurologic signs. The HINTS examination is reassuring.

What does this suggest?

- (A) Multiple sclerosis needing gadolinium MRI
- (B) Peripheral vertigo, with MRI if still unclear
- (C) Vestibular migraine needing a triptan
- (D) Posterior circulation stroke needing thrombolysis
- (E) Cerebellar hemorrhage needing surgery

29 Touching the left cornea causes neither eye to blink, whereas touching the right cornea causes both eyes to blink.

Which nerve is damaged?

- (A) Left CN III
- (B) Right V1
- (C) Right CN VII
- (D) Left V1
- (E) Left CN VII

30 A 64-year-old woman with acute angle closure has an IOP of 62 mmHg. She has received acetazolamide, timolol, and apraclonidine.

When should pilocarpine be added?

- (A) Only if IV mannitol fails
- (B) Once IOP is below 40 mmHg
- (C) Immediately, as the first drug
- (D) Never, because it worsens angle closure
- (E) Only after laser iridotomy

31 A second stimulus delivered 0.5 ms after an action potential produces no response, however strong it is.

Which mechanism explains this?

- (A) Calcium channels are blocked
- (B) The Na/K-ATPase has stopped restoring the gradients
- (C) Potassium leak channels have closed after the spike
- (D) Myelin has been lost
- (E) Voltage-gated sodium channels are inactivated

32 A 60-year-old man with 12 years of Parkinson disease has disabling motor fluctuations and dyskinesia despite optimized levodopa, entacapone, and an agonist. His cognition is normal.

Which treatment is most appropriate?

(A) Adding haloperidol
(B) Tetrabenazine
(C) Stopping levodopa
(D) Adding a second dopamine agonist
(E) DBS

33 A student tests the gag reflex.

Which nerves form its afferent and efferent limbs?

(A) CN X afferent, CN IX efferent
(B) CN IX afferent, CN X efferent
(C) CN IX afferent, CN XII efferent
(D) CN X afferent, CN XI efferent
(E) CN V afferent, CN VII efferent

34 Non-contrast CT 3 hours after sudden aphasia and right hemiparesis is read as showing loss of the gray-white border over the left insula.

What does this finding represent?

(A) A normal variant
(B) An old lacunar infarct
(C) Calcification of the basal ganglia
(D) Subarachnoid hemorrhage
(E) An early sign of ischemia

35 A 68-year-old man with hypertension and glaucoma has sudden painless blurring in his left eye. The fundus shows dilated, tortuous veins with hemorrhages in all four quadrants.

What is the most likely diagnosis?

(A) CRVO
(B) Retinal detachment
(C) Wet macular degeneration
(D) CRAO
(E) Anterior ischemic optic neuropathy

36 A 38-year-old man with flank pain and hematuria has enlarged kidneys full of cysts on ultrasound. His mother died suddenly of a brain hemorrhage at 45.

Which type of brain hemorrhage is he at increased risk of?

(A) Venous infarction from sinus thrombosis
(B) Subdural hemorrhage from bridging veins
(C) Deep hemorrhage from Charcot-Bouchard aneurysms in the pons
(D) Subarachnoid hemorrhage from a saccular aneurysm
(E) Lobar hemorrhage from amyloid angiopathy

37 A 74-year-old man with DLB has troublesome visual hallucinations and worsening attention.

Which drug is most appropriate first?

(A) Amitriptyline
(B) High-dose levodopa
(C) Rivastigmine
(D) Haloperidol
(E) Risperidone

38 A student studies a tract that arises in the red nucleus.

Where does it cross, and which movement does it favor?

(A) Medulla, leg extension
(B) Pons, upper limb extension
(C) Midbrain, upper limb flexion
(D) Spinal cord, axial posture
(E) Uncrossed, antigravity extension

39 A 70-year-old woman with late-onset Alzheimer disease carries two copies of the APOE ε4 allele.

How does APOE ε4 raise her risk?

(A) It forms part of γ-secretase
(B) It increases APP gene dosage
(C) It blocks α-secretase
(D) It slows Aβ clearance
(E) It hyperphosphorylates tau directly

40 A 9-year-old boy has encephalopathy and multifocal deficits 2 weeks after a viral illness. MRI shows fluffy, ill-defined white matter lesions, and serum MOG-IgG is positive at 1:320.

Where is the target antigen of this disease?

(A) Peripheral nerve myelin made by Schwann cells
(B) Water channels on astrocyte end-feet
(C) Voltage-gated calcium channels
(D) The outermost layer of myelin and the oligodendrocyte surface
(E) Nicotinic acetylcholine receptors at the neuromuscular junction

41 A 49-year-old woman had a ruptured aneurysm coiled 6 days ago. Today she becomes confused and develops new right arm weakness. CT head shows no new bleeding.

Which complication is most likely?

(A) Hemorrhagic transformation
(B) Central post-stroke pain
(C) Malignant MCA infarct from the initial bleed
(D) Rebleeding
(E) Vasospasm and delayed cerebral ischemia

42 A researcher wants a CSF test that detects misfolded α-synuclein in a living patient.

Which test is this?

(A) Kappa free light chains
(B) Oligoclonal bands
(C) Seed amplification assay
(D) CSF Aβ42 and tau
(E) CSF Aβ42 with phosphorylated tau

43 A 27-year-old man has painful vision loss in both eyes over a week. Both optic discs are swollen, and MRI shows enhancement of most of each optic nerve, with perineural sheath enhancement.

Which antibody test is most likely to be positive?

(A) MOG-IgG
(B) Anti-acetylcholine receptor
(C) Anti-NMDA receptor
(D) AQP4-IgG
(E) ANA

44 A 79-year-old man taking dabigatran for A-fib has a left lobar hemorrhage on CT. He took his last dose 3 hours ago.

Which reversal agent is most appropriate?

(A) Protamine
(B) IV vitamin K
(C) Idarucizumab
(D) Cryoprecipitate
(E) Andexanet alfa

45 A 70-year-old man falls forward and hits his chin, extending his neck. He has weakness of both arms, worst in his hands, with only mild leg weakness, and loss of pain and temperature sensation over his shoulders.

Which cord syndrome is this?

(A) Complete cord transection
(B) Posterior cord syndrome
(C) Central cord syndrome
(D) Anterior cord syndrome
(E) Brown-Séquard syndrome

46 During a deep breath, slowly adapting stretch receptors in the airways fire increasingly.

What reflex do they mediate?

(A) Reflex bradycardia through the carotid sinus nerve
(B) Bronchoconstriction
(C) Inhibition of inspiration to prevent overinflation
(D) Cough
(E) Faster, shallower breathing in response to interstitial edema

47 A 30-year-old man with a complete C6 spinal cord injury has a sudden pounding headache, flushing above the injury, and a BP of 210/120 mm Hg with HR 48/min.

Which is the most appropriate first step?

(A) Check and relieve bladder distension
(B) Perform carotid sinus massage to slow the heart
(C) Give IV furosemide to lower his blood volume
(D) Give atropine
(E) Lay him flat

48 A 60-year-old hyperopic woman with a shallow anterior chamber takes an over-the-counter cold remedy containing pseudoephedrine. That night she develops a painful red eye with halos.

How did the remedy trigger her symptoms?

- (A) It constricted the pupil, causing pupillary block against the lens
- (B) It clogged the trabecular meshwork with protein
- (C) It swelled the lens through sorbitol
- (D) It decreased aqueous production
- (E) It dilated the pupil, letting the bunched iris narrow the angle

---

49 A 74-year-old woman has a new headache over her temples for 3 weeks, with pain in her jaw when chewing. Her ESR is 88 mm/h.

Which step is most appropriate?

- (A) 100% oxygen during attacks
- (B) Non-contrast CT, then a lumbar puncture
- (C) Stop regular analgesics
- (D) Prompt high-dose glucocorticoids
- (E) Triptan at headache onset

---

50 A 31-year-old woman with relapsing-remitting MS has 5 days of new numbness and weakness of her left leg. Infection has been excluded.

Which is the most appropriate treatment?

- (A) Plasma exchange first
- (B) IVIG
- (C) Start natalizumab today
- (D) Baclofen
- (E) IV methylprednisolone for 3–7 days

TEST A

Answer sheet

Name _____ Date _____ Score _____ / 50

1 A B C D E	11 A B C D E	21 A B C D E	31 A B C D E	41 A B C D E
2 A B C D E	12 A B C D E	22 A B C D E	32 A B C D E	42 A B C D E
3 A B C D E	13 A B C D E	23 A B C D E	33 A B C D E	43 A B C D E
4 A B C D E	14 A B C D E	24 A B C D E	34 A B C D E	44 A B C D E
5 A B C D E	15 A B C D E	25 A B C D E	35 A B C D E	45 A B C D E
6 A B C D E	16 A B C D E	26 A B C D E	36 A B C D E	46 A B C D E
7 A B C D E	17 A B C D E	27 A B C D E	37 A B C D E	47 A B C D E
8 A B C D E	18 A B C D E	28 A B C D E	38 A B C D E	48 A B C D E
9 A B C D E	19 A B C D E	29 A B C D E	39 A B C D E	49 A B C D E
10 A B C D E	20 A B C D E	30 A B C D E	40 A B C D E	50 A B C D E

Fill one bubble for each question. The answer key at the start of the Test A answers uses the same layout.

Test B

50 questions | 60 minutes | Start the timer

1 A 70-year-old man with a COPD exacerbation is given high-flow oxygen, and his $PaCO_2$ rises.

Which explanation is most accepted?

- (A) Ventilation-perfusion mismatch and the Haldane effect
- (B) Oxygen paralyzes the diaphragm
- (C) Loss of his hypoxic drive makes him stop breathing for long periods
- (D) Oxygen blocks the central chemoreceptors on the ventral medulla
- (E) Oxygen stimulates the apneustic center

2 A student notes that smell differs from the other senses in its route to the cortex.

What is distinctive about it?

- (A) It crosses entirely at the chiasm
- (B) It reaches cortex without a thalamic relay
- (C) Its receptor neurons never regenerate once they are lost
- (D) It is carried by the trigeminal nerve
- (E) It relays in the medial geniculate nucleus before the cortex

3 A 39-year-old woman with AQP4-IgG-positive NMOSD is to start eculizumab.

Which step is required before treatment?

- (A) HLA-B*1502 testing
- (B) Checking TPMT activity alone
- (C) Meningococcal vaccination
- (D) Echocardiogram
- (E) JC virus antibody testing

4 A student traces unconscious proprioception from leg muscle spindles to the cerebellum through the uncrossed dorsal tract.

Where does this tract arise, and which peduncle does it enter?

- (A) Red nucleus, inferior cerebellar peduncle
- (B) Spinal border cells, superior cerebellar peduncle
- (C) Dorsal horn, superior cerebellar peduncle
- (D) Clarke nucleus, inferior cerebellar peduncle
- (E) Nucleus gracilis, middle cerebellar peduncle

5 A neurologist notes that her patient with DLB draws a clock poorly and loses track of conversations, yet recalls recent events well.

Which feature of DLB explains the relatively spared memory?

- (A) Lewy bodies in the occipital cortex
- (B) Relative preservation of the medial temporal lobe
- (C) Lewy bodies in the dorsal motor nucleus of the vagus
- (D) Dopamine loss from the substantia nigra
- (E) Loss of the nucleus basalis of Meynert

6 A newborn of hearing parents fails the hearing screen and has severe bilateral SNHL with no other abnormalities. Congenital CMV testing is negative.

Mutations in which gene are most likely?

- (A) EDNRB
- (B) PAX3
- (C) MITF
- (D) The merlin gene
- (E) GJB2

7 A student examines a sensory neuron whose cell body lies in a dorsal root ganglion, with one process that splits into a peripheral and a central branch.

Which type of neuron is this?

- (A) Multipolar, like a motor neuron
- (B) Interneuron
- (C) Pseudounipolar
- (D) Bipolar, with two separate processes
- (E) Bipolar

8 A neurologist proposes testing MOG-IgG in every patient with possible MS.

Why is this discouraged?

(A) The assay requires a brain biopsy
(B) MOGAD is treated the same as MS, so the result changes nothing
(C) MOG-IgG is found in only 0.4% of MS, so false positives result
(D) MOG-IgG is positive in most MS
(E) MOG-IgG predicts disability so strongly that it would alarm patients

9 In early Huntington disease, indirect-pathway striatal neurons die first.

What happens to GPi output and movement?

(A) GPi inhibition rises, causing bradykinesia
(B) STN excitation rises, causing rigidity
(C) Thalamic output falls, causing akinesia
(D) GPi inhibition falls, causing chorea
(E) Dopamine release falls, causing tremor

10 A 71-year-old woman with Parkinson disease has a drug added that prolongs the effect of each levodopa dose.

Which drug and mechanism fit?

(A) Rasagiline, a COMT inhibitor
(B) Entacapone, a COMT inhibitor
(C) Trihexyphenidyl, a dopamine precursor
(D) Amantadine, a D3 receptor agonist
(E) Pramipexole, an MAO-B inhibitor

11 A 34-year-old woman has a stroke the day after a long-haul flight, during which her calf became swollen. Workup finds no A-fib or carotid disease, and a bubble echocardiogram shows a patent foramen ovale.

Which mechanism best explains her stroke?

(A) Global hypoperfusion causing a watershed infarct in the cortex
(B) A venous thrombus crossing a right-to-left shunt
(C) Lipohyalinosis of penetrating arteries
(D) A left atrial thrombus from A-fib
(E) Atherosclerotic thrombus in the ICA

12 A 70-year-old man has parkinsonism that responded moderately to levodopa for several years, with later vertical gaze slowing. His neurologist expects a longer course than classic PSP.

Which PSP subtype fits?

(A) PSP-CBS
(B) PSP-P
(C) PSP-PNFA
(D) PSP-bvFTD
(E) PSP-RS

13 A 48-year-old man has hoarseness, difficulty swallowing, and weakness turning his head to the left and shrugging his right shoulder.

Where is the lesion?

(A) Right jugular foramen
(B) Left medulla
(C) Right cerebellopontine angle
(D) Right C5 root
(E) Right cavernous sinus

14 A 68-year-old man had brief painless loss of vision in his left eye, like a curtain coming down, followed a week later by a minor right hand weakness that resolved. Carotid duplex shows 80% stenosis of the left ICA.

In addition to antiplatelet therapy and a high-intensity statin, which treatment is indicated?

(A) No further treatment
(B) PFO closure
(C) Carotid endarterectomy
(D) Immunosuppression
(E) Anticoagulation with a DOAC

15 An 86-year-old woman has late Alzheimer disease with mutism, incontinence, and difficulty swallowing.

What is the most common cause of death in Alzheimer disease?

(A) Myocardial infarction
(B) Pneumonia
(C) Status epilepticus
(D) Hepatic failure
(E) Lobar ICH

16 An 82-year-old man has a supratentorial hemorrhage estimated at 40 mL on CT, with intraventricular extension. His GCS is 9.

What is his ICH score?

(A) 2
(B) 5
(C) 4
(D) 6
(E) 3

17 A 58-year-old man with bvFTD is started on donepezil by a doctor who assumes it works as in Alzheimer disease.

What is the expected effect?

(A) Prevention of ALS in C9orf72 carriers
(B) Slowed progression of frontal and temporal atrophy
(C) No proven benefit, and behavior may worsen
(D) Improved behavior and cognition within weeks
(E) Reduced hyperorality

18 A 68-year-old woman with early Alzheimer disease and amyloid confirmed on PET starts lecanemab.

Which monitoring does this treatment require?

(A) Lithium levels
(B) ECG for bradycardia only
(C) JC virus status
(D) MRI for ARIA
(E) Liver tests for hepatotoxicity

19 Four months after a stroke, a 63-year-old man has burning pain over his whole right side, worsened by light touch from his clothes. Examination shows reduced sensation on the right but no new deficit.

Which earlier infarct site is most likely?

(A) Left thalamus
(B) Right cerebellar hemisphere
(C) Left frontal eye field
(D) Left occipital cortex
(E) Left caudate

20 A volume load stretches the atria of a healthy volunteer.

What response follows?

(A) Increased sympathetic tone to the kidney
(B) Reduced vagal tone to the heart
(C) Vasopressin release driven by the area postrema
(D) Lower renal sympathetic tone and ANP release, causing natriuresis
(E) Higher RAAS activity and vasopressin release from the pituitary

21 A neurotransmitter binds a postsynaptic receptor and makes the membrane potential more negative.

Which receptor and ion best explain this?

(A) AMPA receptor, sodium entry
(B) NMDA receptor, calcium entry
(C) Voltage-gated calcium channel, calcium entry
(D) Nicotinic receptor, sodium entry
(E) GABA-A receptor, chloride entry

22 A 70-year-old woman sees two overlapping images when she looks at the moon with her right eye alone. Covering the right eye stops it.

What is the most likely cause?

(A) Right optic neuritis
(B) Right CN VI palsy from microvascular ischemia
(C) Left INO
(D) Ocular myasthenia gravis
(E) Cataract in the right eye

23 Two days after SAH, a 60-year-old man becomes increasingly drowsy. CT shows enlarged lateral ventricles.

What is the mechanism of this complication?

(A) The aneurysm compresses the cerebral aqueduct from the start
(B) Hyponatremia causes cerebral edema
(C) Vasospasm infarcts the periventricular white matter
(D) Blood stimulates the choroid plexus to make more CSF
(E) Blood blocks CSF absorption at the arachnoid granulations

24 A 50-year-old man has a thunderclap headache. Non-contrast CT head performed 3 hours after onset is normal, as read by an experienced radiologist.

What is the most appropriate next step?

- (A) SAH can be ruled out without lumbar puncture
- (B) Repeat non-contrast CT in 24 hours
- (C) Catheter angiography
- (D) MRI with gadolinium
- (E) Lumbar puncture for xanthochromia, done at least 12 hours after onset

25 A student compares the lateral and medial spinothalamic pathways.

What does the medial pathway mainly convey?

- (A) Emotional and visceral responses to pain
- (B) Unconscious proprioception
- (C) Fine touch, vibration, and two-point discrimination
- (D) Facial pain only
- (E) Precise location and intensity of the stimulus

26 A 24-year-old woman with poorly controlled type 1 diabetes has blurred vision. Slit lamp examination shows white, flake-like cortical opacities in both lenses.

Which mechanism explains these opacities?

- (A) Urochrome pigment collects in the nucleus
- (B) VEGF drives new vessels from the iris into the lens
- (C) Macrophages clog the trabecular meshwork
- (D) Lens epithelium undergoes fibrous metaplasia at the back of the lens
- (E) Sorbitol accumulates and draws water into the lens

27 A 63-year-old man has gone from normal to unable to dress himself in 3 months, with sudden jerks of his limbs.

Which diagnosis must be considered?

- (A) Alzheimer disease
- (B) Normal pressure hydrocephalus
- (C) Depression
- (D) Prion disease
- (E) Essential tremor

28 A 24-year-old woman who sleeps in her contact lenses has 2 days of a painful red right eye, photophobia, and blurred vision. Slit lamp examination shows a white stromal infiltrate with an overlying epithelial defect.

Which is the most appropriate next step?

- (A) Topical corticosteroids
- (B) Oral acetazolamide
- (C) Artificial tears and cool compresses, with review in a week
- (D) Topical antihistamine drops
- (E) Corneal scraping, then urgent topical antibiotics

29 A 45-year-old woman with NMOSD is starting RTX every 6 months.

Which monitoring is needed?

- (A) Visual evoked potentials yearly
- (B) Serum AQP4-IgG titer before each dose to set the dose
- (C) Immunoglobulin levels before treatment and yearly
- (D) JC virus status every 6 months
- (E) Lymphocyte count to keep it above 0.2 × 10^9/L

30 A 20-month-old boy has a 2-minute generalized seizure during a fever of 39.2 °C from an ear infection. He recovers fully within 30 minutes and has no meningeal signs.

How is this seizure best classified?

- (A) Simple febrile seizure
- (B) Absence seizure
- (C) Status epilepticus
- (D) Epilepsy needing long-term medication
- (E) Complex febrile seizure

31 A tract from the medulla inhibits extensors, facilitates flexors, and contributes to breathing control.

Which tract is this?

- (A) Anterior corticospinal tract
- (B) Pontine reticulospinal tract, uncrossed
- (C) Medullary reticulospinal tract
- (D) Rubrospinal tract
- (E) Lateral vestibulospinal tract from the lateral nucleus

32 A 25-year-old man with drug-resistant focal seizures is being assessed for surgery. MRI is normal.

Which finding on ictal SPECT, injected within 90 seconds of onset, marks the seizure focus?

- (A) Normal symmetric perfusion
- (B) Increased perfusion at the focus
- (C) Reduced striatal uptake
- (D) Amyloid deposition at the focus
- (E) Decreased perfusion at the focus

33 A 50-year-old woman with vestibular neuritis has used meclizine and diazepam daily for 6 weeks and still feels dizzy.

Which is the most appropriate change?

- (A) Perform the Epley maneuver weekly
- (B) Increase the diazepam dose
- (C) Add oral prednisone
- (D) Stop the suppressants and start vestibular rehabilitation
- (E) Refer for intratympanic gentamicin to ablate the labyrinth

34 A 76-year-old man struggles to follow conversation in restaurants and mishears words like "fit" and "sit." Both eardrums are normal.

Which audiogram pattern is most likely?

- (A) A notch at 4 kHz with recovery at 8 kHz
- (B) An air-bone gap in both ears
- (C) Bilateral, high-frequency, downsloping SNHL
- (D) Low-frequency SNHL in one ear that fluctuates
- (E) A flat loss in one ear with an air-bone gap in the other

35 A 52-year-old woman has a tremor of both hands when holding a cup or writing, which improves with a glass of wine. Her father had the same tremor. There is no bradykinesia.

Which drug is most appropriate?

- (A) Trihexyphenidyl
- (B) Haloperidol
- (C) Levodopa
- (D) Tetrabenazine
- (E) Propranolol

36 A 62-year-old East Asian woman has sudden severe pain in her right eye, halos around lights, headache, and vomiting after an evening in a dark cinema. The eye is red and firm, the cornea hazy, and the pupil fixed and mid-dilated. IOP is 58 mmHg.

After lowering the IOP medically, which treatment is definitive?

- (A) Topical latanoprost long term
- (B) Topical corticosteroids
- (C) Topical atropine
- (D) Laser peripheral iridotomy
- (E) Laser trabeculoplasty

37 A 76-year-old man with hypertension, diabetes, and two TIAs has had a stepwise decline in thinking over 2 years, with sudden drops followed by plateaus.

Which diagnosis is most likely?

- (A) Alzheimer disease
- (B) Vascular dementia
- (C) NPH
- (D) Prion disease
- (E) Frontotemporal dementia

38 Massage over the carotid sinus slows a patient's heart rate.

Which nerve carries the afferent signal?

- (A) CN VII
- (B) CN X
- (C) CN XII
- (D) CN XI
- (E) CN IX

39 A 45-year-old woman had relapsing-remitting MS from age 30. Over the last 3 years her walking has gradually worsened, without any relapses.

Which course does this describe?

- (A) Clinically isolated syndrome
- (B) Radiologically isolated syndrome
- (C) Relapsing-remitting
- (D) Secondary progressive
- (E) Primary progressive

40 A 25-year-old woman has had a severe asthma attack for 6 hours. Her PaCO2, low on arrival, is now 42 mmHg.

What does this normal PaCO2 indicate?

(A) She has metabolic alkalosis
(B) Peripheral chemoreceptors are suppressed
(C) The attack is resolving
(D) Her central drive has failed
(E) Her respiratory muscles are tiring

41 A 40-year-old man on haloperidol develops slowness and rigidity.

Which receptor blockade explains this?

(A) GABA-A receptors on neurons of the GPi
(B) Muscarinic receptors in the cortex
(C) D1 receptors on direct-pathway neurons in the striatum
(D) NMDA receptors in the STN
(E) D2 receptors on indirect-pathway neurons

42 A 50-year-old man has right facial weakness after temporal bone surgery, with a dry right eye but normal taste and hearing.

Which branch's fibers explain the dry eye?

(A) Greater petrosal nerve to the lacrimal gland
(B) Ophthalmic division of CN V
(C) Branches within the parotid gland
(D) Nerve to stapedius
(E) Chorda tympani to the submandibular and sublingual glands

43 A student notes that CN IV is unusual among the cranial nerves.

Which feature makes it unusual?

(A) It runs free within the cavernous sinus rather than in its wall
(B) Every fiber crosses before leaving the back of the midbrain
(C) It carries parasympathetic fibers
(D) It leaves through the interpeduncular fossa beside the PComm
(E) Its nucleus is the horizontal gaze center

44 A 45-year-old man with epilepsy is found after a witnessed tonic-clonic seizure with a bitten tongue. An hour later he has left arm weakness that is improving. CT head is normal and glucose is normal.

Which is the most appropriate management of his weakness?

(A) Observation, expecting it to resolve
(B) IV methylprednisolone
(C) Start aspirin and clopidogrel
(D) CT angiogram and mechanical thrombectomy
(E) IV alteplase

45 A 60-year-old smoker has wasting and clawing of his left hand, numbness of his medial forearm, and a small left pupil with ptosis.

Which lesion best explains these findings?

(A) Median nerve in the carpal tunnel
(B) Ulnar nerve at the cubital tunnel
(C) C8–T1 injury from an apical lung tumor
(D) Left lateral medullary stroke from PICA occlusion
(E) C5–C6 traction injury of the upper trunk

46 MRI of a woman with old MS plaques shows a lesion bright on DWI. The same lesion is bright on ADC.

What does this combination indicate?

(A) Acute infarct under 4.5 hours old
(B) T2 shine-through, not acute infarct
(C) An abscess
(D) Acute hemorrhage
(E) An active plaque that would enhance with gadolinium

47 A neurophysiologist records inhibitory transmission onto motor neurons in the spinal cord.

Which transmitter is inhibitory mainly in the cord?

(A) Glutamate
(B) Dopamine
(C) Acetylcholine
(D) Glycine
(E) Norepinephrine

48 A 67-year-old woman has fallen backward several times in the last year. She has a stiff, upright posture and cannot look down voluntarily, although her eyes move down when her head is tilted back.

Which diagnosis is most likely?

(A) Corticobasal degeneration
(B) Dementia with Lewy bodies
(C) Multiple system atrophy
(D) Progressive supranuclear palsy
(E) Typical Parkinson disease

---

49 A 42-year-old man has restless, flowing, involuntary movements of his fingers, face, and trunk, with irritability and poor planning.

Loss of which neurons first explains his movements?

(A) Indirect-pathway medium spiny neurons projecting to the GPe
(B) Direct-pathway D1 medium spiny neurons projecting to the GPi
(C) Dopamine neurons of the SNc
(D) Cholinergic neurons of the nucleus basalis
(E) Purkinje cells of the cerebellum

---

50 A 66-year-old woman has frequent word-finding pauses and cannot repeat long sentences, but her grammar and word meanings are intact.

Which underlying disease is most likely?

(A) Alzheimer disease
(B) Huntington disease
(C) Dementia with Lewy bodies
(D) FTLD-TDP
(E) FTLD-tau with Pick bodies

TEST B

Answer sheet

Name _____ Date _____ Score _____ / 50

1 A B C D E	11 A B C D E	21 A B C D E	31 A B C D E	41 A B C D E
2 A B C D E	12 A B C D E	22 A B C D E	32 A B C D E	42 A B C D E
3 A B C D E	13 A B C D E	23 A B C D E	33 A B C D E	43 A B C D E
4 A B C D E	14 A B C D E	24 A B C D E	34 A B C D E	44 A B C D E
5 A B C D E	15 A B C D E	25 A B C D E	35 A B C D E	45 A B C D E
6 A B C D E	16 A B C D E	26 A B C D E	36 A B C D E	46 A B C D E
7 A B C D E	17 A B C D E	27 A B C D E	37 A B C D E	47 A B C D E
8 A B C D E	18 A B C D E	28 A B C D E	38 A B C D E	48 A B C D E
9 A B C D E	19 A B C D E	29 A B C D E	39 A B C D E	49 A B C D E
10 A B C D E	20 A B C D E	30 A B C D E	40 A B C D E	50 A B C D E

Fill one bubble for each question. The answer key at the start of the Test B answers uses the same layout.

Test C

50 questions | 60 minutes | Start the timer

1 A 12-year-old boy whose father has Huntington disease has rigidity, slowness, dystonia, falling school grades, and seizures.

Which form of the disease is this?

- (A) Westphal variant of juvenile Huntington disease
- (B) Wilson disease
- (C) Chorea-acanthocytosis
- (D) Late-onset Huntington disease with a mild course
- (E) McLeod syndrome

2 A 26-year-old woman with highly active relapsing MS is to start natalizumab.

Which test should be checked first?

- (A) JC virus status
- (B) AQP4 antibody
- (C) Thiopurine methyltransferase
- (D) HLA-B*1502
- (E) First-dose ECG monitoring

3 A student identifies the output nuclei of the basal ganglia.

What is their resting effect on the thalamus?

- (A) The GPe tonically excites the VA thalamus
- (B) The GPi and SNr tonically inhibit the VA and VL thalamus
- (C) The SNc tonically inhibits the VL thalamus
- (D) The striatum tonically excites the VA and VL thalamus through glutamate
- (E) The STN tonically inhibits the thalamus

4 A 74-year-old man suddenly develops violent, flinging movements of his left arm and leg.

Where is the lesion most likely?

- (A) Right subthalamic nucleus
- (B) Right cerebellar hemisphere
- (C) Left caudate
- (D) Right substantia nigra
- (E) Left subthalamic nucleus

5 A 30-year-old woman has drooping eyelids and double vision that worsen through the day, with fatigable arm weakness.

Which is the first-line symptomatic drug?

- (A) Sertraline
- (B) Pyridostigmine
- (C) Lorazepam
- (D) Atropine
- (E) Botulinum toxin

6 An 83-year-old man with a hypermature cataract has a painful red eye with IOP of 44 mmHg. Gonioscopy shows an open angle, and white cells float in the anterior chamber.

What is the mechanism of his raised IOP?

- (A) A swollen, intumescent lens pushes the iris forward and closes the angle
- (B) Pupillary block from a dilated pupil
- (C) Neovascularization of the angle
- (D) Lens proteins leak and macrophages clog the trabecular meshwork
- (E) Steroid-induced reduction in outflow through the meshwork

7 A man's tongue deviates to the left on protrusion, with wasting and fasciculations of the left side.

Which nerve is damaged?

- (A) Left CN IX
- (B) Left CN XII
- (C) Right corticobulbar tract
- (D) Left CN X
- (E) Right CN XII

8 A tumor at the skull base causes numbness of the lower lip and chin and weakness of the masseter on one side.

Through which foramen does the affected division pass?

- (A) Jugular foramen
- (B) Foramen rotundum
- (C) Stylomastoid foramen
- (D) Superior orbital fissure
- (E) Foramen ovale

9 A 70-year-old man notices his eyelashes have grown longer and his light brown iris has darkened since starting a glaucoma drop.

How does this drop lower IOP?

(A) Constricts ciliary body vessels through α1 receptors
(B) Reduces bicarbonate in the ciliary body
(C) Contracts the ciliary muscle to open the trabecular meshwork
(D) Decreases aqueous production by blocking ciliary β receptors
(E) Increases uveoscleral outflow

10 A 30-year-old woman has a first attack of myelitis. CSF shows 85 white cells/μL with neutrophils and eosinophils and no oligoclonal bands.

Which diagnosis do these CSF findings favor?

(A) Small vessel disease
(B) Functional neurologic disorder
(C) Multiple sclerosis
(D) NMOSD
(E) B12 deficiency

11 A 60-year-old man has small, irregular pupils that constrict when he looks at a near target but not to light.

Which condition do these pupils mark?

(A) Neurosyphilis
(B) CN III palsy
(C) Opioid toxicity
(D) Optic neuritis
(E) Horner syndrome

12 A 70-year-old woman has a stroke in the right medial medulla.

Which sensory loss is expected?

(A) Left-sided pain and temperature loss
(B) Cape-like pain and temperature loss
(C) Left-sided vibration and position loss
(D) Right-sided ataxia
(E) Right-sided vibration and position loss

13 An 84-year-old woman in a nursing home became confused over 2 days. Her attention wanders, and she is drowsy in the morning but agitated at night. She has a fever and cloudy urine.

Which is the most likely cause of her confusion?

(A) Delirium
(B) Alzheimer disease
(C) Frontotemporal dementia
(D) Vascular dementia
(E) Normal pressure hydrocephalus

14 A student traces vibration sense from the big toe up the spinal cord.

In which fasciculus does it ascend?

(A) Lateral spinothalamic tract
(B) Fasciculus cuneatus
(C) Dorsal spinocerebellar tract
(D) Anterior spinothalamic tract
(E) Fasciculus gracilis

15 A student traces baroreceptor afferents to their cell bodies.

Where are the cell bodies of the carotid and aortic afferents?

(A) Petrosal and nodose ganglia
(B) Nodose and petrosal ganglia
(C) Superior cervical and stellate ganglia
(D) Geniculate and trigeminal ganglia
(E) Dorsal root ganglia at T1–T4

16 A 40-year-old woman has twitching that starts in her left thumb, spreads over a minute to her hand and arm, then to the left side of her face.

What does this spread indicate?

(A) A migraine aura in the occipital cortex
(B) A spinal cord lesion
(C) Transient ischemia of the internal capsule
(D) Peripheral nerve hyperexcitability
(E) A seizure spreading along the motor cortex

17 A 70-year-old woman presents 7 hours after onset of left hemiparesis. CT shows no hemorrhage and CT angiogram shows no large vessel occlusion. She will not receive reperfusion therapy. Her BP is 196/104 mm Hg.

Which is the most appropriate approach to her BP?

- (A) Lower it immediately to below 140/90
- (B) Allow it up to 220/120
- (C) Give IV nicardipine to a systolic target of 140
- (D) Lower it to below 185/110 for thrombolysis
- (E) Give nimodipine

18 After a head injury in a cycling crash, a 19-year-old woman has vertical double vision, worse reading and going downstairs, and tilts her head away from the affected eye.

Which nerve is most likely affected?

- (A) CN VI
- (B) CN IV
- (C) CN III
- (D) MLF
- (E) CN VII

19 A 38-year-old woman has severe bilateral optic neuritis and a cord lesion spanning 5 vertebral segments. Serum AQP4-IgG is positive.

Which cell is the primary target of the disease?

- (A) Motor neuron
- (B) Oligodendrocyte
- (C) Schwann cell
- (D) Astrocyte
- (E) Microglia

20 A 19-year-old woman has an episode of thrashing with her eyes closed, side-to-side head movement, and asynchronous limb movements for 4 minutes. She is alert immediately afterward.

Which test best confirms the diagnosis?

- (A) Lumbar puncture
- (B) Toxicology screen
- (C) Video EEG
- (D) Serum prolactin 10–20 minutes later
- (E) MRI brain

21 A 4-year-old boy wakes with both eyelids stuck together by yellow discharge. His eyes are red, but his vision is normal.

Which treatment shortens this illness?

- (A) Oral aciclovir
- (B) Topical antihistamines and mast cell stabilizers
- (C) Topical corticosteroids to reduce the inflammation
- (D) Intravitreal anti-VEGF
- (E) Topical antibiotic drops such as chloramphenicol

22 Six weeks after a wrist fracture, a 48-year-old woman has severe burning pain in her hand, pain from light touch, and a hand that is swollen, sweaty, and changes color.

What is the most likely diagnosis?

- (A) Raynaud phenomenon
- (B) Gout
- (C) CRPS
- (D) Carpal tunnel syndrome
- (E) Cervical radiculopathy

23 A 45-year-old man taking a long-standing antiseizure drug has swollen, overgrown gums. His dose was recently increased, and he now has nystagmus and an unsteady gait, with a level far higher than the dose increase predicted.

Which drug is he most likely taking?

- (A) Levetiracetam
- (B) Phenytoin
- (C) Lamotrigine
- (D) Valproate
- (E) Carbamazepine

24 A 21-year-old woman collapsed after standing for an hour at a hot concert. She felt lightheaded and her vision greyed first, had a few brief jerks, and was fully alert within a minute.

Which tests are most appropriate?

- (A) Video EEG
- (B) MRI brain with gadolinium
- (C) ECG and orthostatic BP
- (D) DAT SPECT
- (E) Lumbar puncture

25 An 80-year-old man with DLB has distressing psychosis despite rivastigmine, and his team decides an antipsychotic is essential.

Which choice is most appropriate?

- (A) Low-dose quetiapine
- (B) Haloperidol at a low dose
- (C) Aripiprazole
- (D) Olanzapine
- (E) Risperidone at night

26 A man with a left cortical stroke has right lower facial weakness but can raise both eyebrows.

Why is the forehead spared?

- (A) The forehead muscles are supplied by the ophthalmic division of the trigeminal nerve
- (B) The upper face receives corticobulbar input from both hemispheres
- (C) The forehead muscles are supplied by the rubrospinal tract
- (D) Forehead fibers cross twice
- (E) The facial nucleus for the forehead lies lower, in the medulla

27 A 46-year-old man with an unsecured ruptured AComm aneurysm has BP 182/100 mm Hg while awaiting coiling.

Which BP target is most appropriate until the aneurysm is secured?

- (A) Systolic below 160 with IV nicardipine or labetalol
- (B) No treatment unless BP rises above 220/120 mm Hg
- (C) Systolic below 185 for thrombolysis
- (D) Systolic below 110 with nitroprusside
- (E) Below 130/80 mm Hg with oral agents started at once

28 A 60-year-old man no longer recognizes the faces of his friends and gets lost in his own neighborhood. MRI shows asymmetric atrophy.

Which region is most affected?

- (A) Left inferior frontal gyrus
- (B) Cerebellar vermis
- (C) Left anterior temporal lobe
- (D) Right anterior temporal lobe
- (E) Dorsolateral prefrontal cortex

29 A 41-year-old man has myelitis with a conus lesion. Serum MOG-IgG on a live cell-based assay is positive at 1:40, and AQP4-IgG is negative.

Under the 2023 criteria, can MOGAD be diagnosed?

- (A) Yes, the conus lesion is a supporting feature
- (B) No, myelitis is not a MOGAD syndrome
- (C) No, a titer of 1:100 or more is always required
- (D) Yes, without needing to exclude MS
- (E) No, MOG-IgG must also be found in CSF

30 A 52-year-old woman with a BMI of 48 has daytime sleepiness and a chronically raised $PaCO_2$ with a high bicarbonate.

Why does her hypercapnia persist?

- (A) She has metabolic acidosis
- (B) Her lung stretch receptors fire continuously and inhibit inspiration
- (C) Retained bicarbonate blunts the central chemoreceptors
- (D) Her preBötzinger complex is absent
- (E) Her carotid bodies have been destroyed by chronic hypoxia

31 A dentist injects lidocaine before filling a tooth, and the patient feels no pain.

What is the mechanism?

- (A) Inhibits the Na/K-ATPase
- (B) Activates GABA-A receptors
- (C) Opens potassium leak channels
- (D) Blocks voltage-gated sodium channels in the nerve
- (E) Blocks voltage-gated calcium channels at the terminal

32 A 62-year-old woman has horizontal double vision from a right abduction deficit, weakness of the whole right side of her face including the forehead, and weakness of her left arm and leg.

Which syndrome is this?

- (A) Lateral pontine (AICA) syndrome
- (B) Inferior medial pontine (Foville) syndrome
- (C) Lateral medullary syndrome
- (D) Ventral pontine (Millard-Gubler) syndrome
- (E) Weber syndrome

33 A 74-year-old smoker has slowly worsening central vision in both eyes over years. The macula shows yellow deposits and pigment changes, and OCT shows no fluid.

Which treatment slows progression?

- (A) Pneumatic retinopexy
- (B) Intravitreal ranibizumab every month
- (C) Antioxidant and zinc supplements
- (D) Laser trabeculoplasty
- (E) IV methylprednisolone followed by an oral taper

34 A 72-year-old man with Parkinson disease feels faint on standing. His BP is 138/82 lying and 112/70 mm Hg after 2 minutes standing, with little change in HR.

How is this best described?

- (A) POTS
- (B) Normal postural change
- (C) Carotid sinus hypersensitivity
- (D) Vasovagal syncope
- (E) Orthostatic hypotension

35 MRI of a 40-year-old woman with a family history of chorea shows atrophy of both caudate nuclei, with enlarged frontal horns. She has no symptoms yet.

Which condition do these findings suggest?

- (A) Parkinson disease
- (B) Mesial temporal lobe epilepsy
- (C) Progressive supranuclear palsy
- (D) Huntington disease
- (E) Multiple system atrophy

36 A 35-year-old man has 2 days of continuous vertigo after a cold, with hearing loss and ringing in his right ear. Nystagmus is unidirectional and suppressed by fixation.

What is the most likely diagnosis?

- (A) BPPV
- (B) Vestibular neuritis
- (C) Labyrinthitis
- (D) Ménière disease
- (E) Vestibular migraine

37 A 45-year-old man has had 3 attacks of vertigo lasting 2–4 hours, each with roaring tinnitus and fullness in his left ear. Audiometry shows left low-frequency sensorineural loss.

Which is the most appropriate first treatment?

- (A) Surgical section of the vestibular nerve as first-line treatment
- (B) Intratympanic gentamicin
- (C) The Epley maneuver
- (D) Long-term daily meclizine
- (E) Less salt, caffeine, and alcohol, with a thiazide or betahistine

38 A 55-year-old man treated for a melanoma 2 years ago has a seizure. CT shows a lobar hemorrhage, and MRI with contrast shows an enhancing mass at its edge.

What is the most likely cause of the hemorrhage?

- (A) Cerebral venous sinus thrombosis
- (B) Mycotic aneurysm
- (C) Hypertensive hemorrhage
- (D) Cerebral amyloid angiopathy
- (E) Hemorrhagic metastasis

39 A man has total deafness in his right ear after a stroke.

Why is a cortical stroke unlikely to be the cause?

- (A) The cochlear nerve does not reach the brainstem
- (B) The auditory cortex lies in the occipital lobe
- (C) Hearing becomes bilateral above the cochlear nuclei
- (D) The auditory cortex receives input only from the opposite ear
- (E) Hearing is carried by CN VII

40 A 35-year-old man who often sits with his legs crossed has a right foot drop. Dorsiflexion and eversion are weak, the dorsum of the foot is numb, but inversion is strong.

Which nerve is injured?

- (A) Femoral nerve
- (B) Sciatic nerve
- (C) Tibial nerve
- (D) Common peroneal nerve at the fibular head
- (E) L5 root

41 A 69-year-old woman who has taken levodopa-carbidopa for 6 years notices her tremor and slowness return about 2 hours before each dose is due.

Which is the most appropriate change?

(A) Stop levodopa
(B) Add trihexyphenidyl only
(C) Add entacapone
(D) Add haloperidol
(E) Switch to amantadine alone

42 A 28-year-old woman has optic neuritis. MRI shows T2 plaques in periventricular and juxtacortical sites, all non-enhancing. She has had no previous symptoms.

Under the 2024 McDonald criteria, which result would most directly confirm MS now?

(A) Positive AQP4 antibody
(B) CSF oligoclonal bands
(C) A raised ESR
(D) Visual evoked potentials showing slowed conduction
(E) A normal B12 level

43 A 3-year-old boy has speech delay. Otoscopy shows dull, retracted eardrums with fluid behind them in both ears, persisting for 4 months. Audiometry shows conductive loss.

Which treatment is most appropriate?

(A) GJB2 testing alone
(B) Stapes surgery
(C) Oral prednisone
(D) Cochlear implants
(E) Ventilation tubes

44 A 70-year-old man with Parkinson disease starts levodopa combined with carbidopa.

What is the role of carbidopa?

(A) It stimulates D3 receptors directly
(B) It blocks peripheral conversion of levodopa to dopamine
(C) It blocks NMDA receptors
(D) It inhibits MAO-B in the brain
(E) It inhibits COMT in the brain to prolong each dose

45 A 55-year-old man with diabetes has a left eye that is down and out with ptosis, but the pupil is normal.

Why is the pupil spared?

(A) Pupil fibers travel in CN IV
(B) Ischemia affects the core of the nerve, sparing the surface pupil fibers
(C) The lesion lies in the Edinger-Westphal nucleus of the midbrain
(D) The pupil is controlled only by sympathetic fibers from T1
(E) Diabetes protects parasympathetic fibers

46 A 70-year-old right-handed woman suddenly cannot read her newspaper, although she can still write a letter. She has a right homonymous hemianopia with macular sparing and no weakness.

Which artery is most likely occluded?

(A) Left posterior cerebral artery
(B) Left anterior cerebral artery
(C) Left middle cerebral artery
(D) Left ophthalmic artery
(E) Right posterior cerebral artery

47 A 58-year-old man with 20 years of poorly controlled hypertension has a pontine hemorrhage.

Which vascular change most likely underlies the bleed?

(A) Saccular aneurysm at a bifurcation of the circle of Willis
(B) A tangle of arteries draining directly into veins
(C) Septic emboli lodging in distal arteries
(D) Amyloid deposits in cortical vessel walls
(E) Charcot-Bouchard aneurysms in penetrating arterioles

48 A radiologist wants a sequence that makes lesions bright while keeping CSF dark, to show plaques next to the ventricles.

Which sequence is this?

(A) T1 without contrast
(B) DWI
(C) MRA
(D) FLAIR
(E) SWI

49 At autopsy, a patient with parkinsonism, early autonomic failure, and ataxia has α-synuclein inclusions in oligodendrocytes.

Which disease is this?

- (A) Typical Parkinson disease
- (B) Progressive supranuclear palsy
- (C) Multiple system atrophy
- (D) Corticobasal degeneration
- (E) Frontotemporal lobar degeneration

---

50 A 50-year-old woman with Huntington disease has chorea that makes her drop things and fall. She has a history of depression.

If tetrabenazine is started, what must be monitored?

- (A) Serum lithium level
- (B) Liver enzymes weekly
- (C) Serum potassium
- (D) Mood and suicidal thoughts
- (E) White cell count for agranulocytosis

TEST C

Answer sheet

Name _____ Date _____ Score _____ / 50

1 (A) (B) (C) (D) (E)	11 (A) (B) (C) (D) (E)	21 (A) (B) (C) (D) (E)	31 (A) (B) (C) (D) (E)	41 (A) (B) (C) (D) (E)
2 (A) (B) (C) (D) (E)	12 (A) (B) (C) (D) (E)	22 (A) (B) (C) (D) (E)	32 (A) (B) (C) (D) (E)	42 (A) (B) (C) (D) (E)
3 (A) (B) (C) (D) (E)	13 (A) (B) (C) (D) (E)	23 (A) (B) (C) (D) (E)	33 (A) (B) (C) (D) (E)	43 (A) (B) (C) (D) (E)
4 (A) (B) (C) (D) (E)	14 (A) (B) (C) (D) (E)	24 (A) (B) (C) (D) (E)	34 (A) (B) (C) (D) (E)	44 (A) (B) (C) (D) (E)
5 (A) (B) (C) (D) (E)	15 (A) (B) (C) (D) (E)	25 (A) (B) (C) (D) (E)	35 (A) (B) (C) (D) (E)	45 (A) (B) (C) (D) (E)
6 (A) (B) (C) (D) (E)	16 (A) (B) (C) (D) (E)	26 (A) (B) (C) (D) (E)	36 (A) (B) (C) (D) (E)	46 (A) (B) (C) (D) (E)
7 (A) (B) (C) (D) (E)	17 (A) (B) (C) (D) (E)	27 (A) (B) (C) (D) (E)	37 (A) (B) (C) (D) (E)	47 (A) (B) (C) (D) (E)
8 (A) (B) (C) (D) (E)	18 (A) (B) (C) (D) (E)	28 (A) (B) (C) (D) (E)	38 (A) (B) (C) (D) (E)	48 (A) (B) (C) (D) (E)
9 (A) (B) (C) (D) (E)	19 (A) (B) (C) (D) (E)	29 (A) (B) (C) (D) (E)	39 (A) (B) (C) (D) (E)	49 (A) (B) (C) (D) (E)
10 (A) (B) (C) (D) (E)	20 (A) (B) (C) (D) (E)	30 (A) (B) (C) (D) (E)	40 (A) (B) (C) (D) (E)	50 (A) (B) (C) (D) (E)

Fill one bubble for each question. The answer key at the start of the Test C answers uses the same layout.

Test A answers

Answer key | Explanations

1 D	11 D	21 E	31 E	41 E
2 B	12 A	22 D	32 E	42 C
3 B	13 E	23 C	33 B	43 A
4 D	14 E	24 B	34 E	44 C
5 A	15 A	25 C	35 A	45 C
6 A	16 B	26 D	36 D	46 C
7 C	17 A	27 A	37 C	47 A
8 A	18 B	28 B	38 C	48 E
9 B	19 E	29 D	39 D	49 D
10 C	20 B	30 B	40 D	50 E

1 **D** **Decreased Aβ42 with increased tau and phosphorylated tau**

Alzheimer disease | Step 2

CSF in Alzheimer disease shows decreased Aβ42 with increased tau and phosphorylated tau, and amyloid PET shows plaques in life.

2 **B** **Uhthoff phenomenon**

Multiple sclerosis | Step 1

In Uhthoff phenomenon, symptoms worsen with heat. It is managed by avoiding heat and is not a relapse, which develops over days to weeks.

WHY NOT THE OTHERS

(A) Lhermitte sign is an electric shock down the spine on neck flexion.

(C) A relapse develops over days to weeks and does not settle within an hour.

3 **B** **To prevent amblyopia**

Cataract | Step 2

Congenital cataracts follow genetic causes, intrauterine infection, and metabolic disease such as galactosemia, present with leukocoria, and are treated early because they cause amblyopia.

4 **D** **Posterior limb**

Physiology | Descending tracts | Step 1

The lateral corticospinal tract descends through the posterior limb of the internal capsule, whereas the corticobulbar tract passes through the genu.

WHY NOT THE OTHERS

(C) The genu carries the corticobulbar tract.

5 **A** **Opening potassium channels and closing N-type calcium channels**

Physiology | Pain pathways and descending modulation | Step 1

Opioid receptors are Gi-coupled: they open potassium channels to hyperpolarize the neuron and close N-type calcium channels to cut transmitter release, such as substance P.

6 **A** **Treat the cause and avoid fluid restriction**

Subarachnoid hemorrhage | Step 2

Hyponatremia after SAH follows SIADH or cerebral salt wasting. It is treated by treating the cause and avoiding fluid restriction, since vasospasm is treated with euvolemia.

WHY NOT THE OTHERS

(C) Fluid restriction risks hypovolemia during the vasospasm window.

7 **C** **Chorea-acanthocytosis**

Huntington disease · Step 2

Chorea-acanthocytosis, autosomal recessive, causes orofacial chorea worse with eating, tongue biting, myopathy, and onset around 30, with acanthocytes on the blood film.

WHY NOT THE OTHERS

- (B) His HTT repeat length is normal.
- (D) Spinocerebellar ataxias cause prominent ataxia.
- (E) Wilson disease causes orofacial dystonia with parkinsonism.

8 **A** **MRI of the brain and orbits with gadolinium**

Vision loss and red eye · Step 2

Optic neuritis causes painful monocular vision loss with an RAPD and reduced color vision, and is often the first sign of MS, so MRI of the brain and orbits is obtained. IV methylprednisolone speeds recovery without changing the outcome.

WHY NOT THE OTHERS

- (E) Temporal artery biopsy is for GCA, after age 50.

9 **B** **Right optic tract**

Physiology · I, II, and VIII · Step 1

Each optic tract carries the opposite visual field to the LGN, so tract and V1 lesions cause homonymous hemianopia.

WHY NOT THE OTHERS

- (C) A Meyer loop lesion causes an upper quadrantanopia.
- (D) A left tract lesion causes a right hemianopia.
- (E) Chiasmal lesions cause bitemporal hemianopia.

10 **C** **Presynaptic voltage-gated calcium channels**

Physiology · Synapses and neurotransmitters · Step 2

Lambert-Eaton syndrome targets presynaptic voltage-gated calcium channels, reducing ACh release, and causes proximal weakness, autonomic dysfunction, and reduced reflexes that improve with repeated contraction, often with small cell lung cancer.

WHY NOT THE OTHERS

- (E) Myasthenia gravis targets postsynaptic AChR and causes fatigable weakness.

11 **D** **Temporal lobe**

Seizures · Step 1

Focal autonomic and psychological seizures arise in the temporal lobe and insula, with auras such as déjà vu and nausea. Involvement of the reticular formation or thalamus brings LOC, automatisms such as chewing and lip smacking, and a post-ictal state.

WHY NOT THE OTHERS

- (A) Frontal lobe seizures are motor.
- (E) An aura shows a focal onset; absence seizures have no post-ictal state.

12 **A** **Dorsal columns**

Motor signs and syndromes · Step 1

Sensory ataxia follows dorsal column or large-fiber loss, with a stomping gait and a positive Romberg test, as in B12 deficiency. Cerebellar ataxia persists with the eyes open.

13 **E** **3 sodium out and 2 potassium in**

Physiology · Neurons and action potentials · Step 1

The Na/K-ATPase pumps 3 sodium out and 2 potassium in to maintain the gradients.

14 **E** **Multiple sclerosis**

NMOSD · Step 1

Short, peripheral cord lesions, Dawson fingers, OCB, and a progressive course favor MS; OCB are usually absent in NMOSD and their presence is a red flag.

WHY NOT THE OTHERS

- (A) MOGAD favors disc edema, conus lesions, and fluffy lesions.
- (D) NMOSD usually lacks OCB and has long central cord lesions.

15 **A** **Left anterior temporal lobe**

Frontotemporal dementia · Step 1

The semantic variant follows left anterior temporal atrophy and causes fluent speech that has lost meaning, with anomia and poor single-word comprehension. The nonfluent variant follows left inferior frontal and insular atrophy.

WHY NOT THE OTHERS

- (B) Right anterior temporal atrophy causes prosopagnosia and getting lost.
- (C) Left inferior frontal atrophy causes effortful, agrammatic speech.

16 **B** **Cerebral amyloid angiopathy**

Intracerebral hemorrhage · Step 1

Cerebral amyloid angiopathy is a common cause of lobar bleeds in older adults, may present with

dementia, and shows lobar microbleeds on MRI with SWI. Hypertension bleeds in deep structures such as the basal ganglia, thalamus, pons, and cerebellum.

WHY NOT THE OTHERS

- (C) AVMs are a cause in young patients.
- (D) Hypertensive bleeds are deep, and she has no hypertension.

---

17 **(A) It increases, inhibiting the thalamus and causing bradykinesia**

Physiology · Basal ganglia pathways · Step 1

Loss of SNc dopamine weakens the direct pathway and strengthens the indirect pathway, increasing GPi inhibition of the thalamus and causing bradykinesia.

---

18 **(B) Pain and temperature in the left leg**

Physiology · Ascending tracts · Step 1

In Brown-Séquard syndrome, the dorsal columns and corticospinal tract are lost on the same side, and spinothalamic fibers, which cross within 1–2 segments of entry, are lost on the opposite side.

---

19 **(E) Left CN VI nucleus**

Physiology · III, IV, and VI · Step 2

The CN VI nucleus is the horizontal gaze center, linked through the MLF to the opposite CN III nucleus, so its lesion abolishes gaze to that side in both eyes.

WHY NOT THE OTHERS

- (A) An MLF lesion causes INO with only adduction failure.
- (B) A nerve lesion weakens only the left lateral rectus.

---

20 **(B) Oral prednisone 1 mg/kg for 7 days, then tapered**

Hearing loss · Step 2

Sudden SNHL, a loss of over 30 dB at 3 consecutive frequencies within 72 hours, is an emergency treated with oral prednisone, because spontaneous recovery is most likely in the first 2 weeks. MRI of the internal auditory canals is also obtained.

---

21 **(E) Saccule**

Vertigo · Step 1

The utricle and saccule sense linear acceleration, gravity, and head tilt, the utricle horizontal and the saccule vertical motion; the canals sense rotation.

WHY NOT THE OTHERS

- (A) The horizontal canal senses yaw.
- (B) The utricle senses horizontal linear acceleration.
- (D) The cochlea transduces sound.

---

22 **(D) Duloxetine**

Physiology · Pain pathways and descending modulation · Step 2

SNRIs and TCAs, such as duloxetine and amitriptyline, block serotonin and norepinephrine reuptake in the spinal cord, strengthening descending inhibition and relieving neuropathic pain.

---

23 **(C) PSP, Alzheimer disease, or other pathologies can cause it**

Atypical parkinsonism · Step 1

CBS is not specific, since PSP, AD, and other pathologies can cause it, and only autopsy confirms the protein.

WHY NOT THE OTHERS

- (A) DAT SPECT is reduced in all three and does not identify the protein.
- (B) CBD is a 4R tauopathy.
- (E) The hot cross bun sign suggests MSA.

---

24 **(B) Glucose and non-contrast CT head**

Acute ischemic stroke · Step 2

Glucose and non-contrast CT head come first, to rule out hypoglycemia and hemorrhage before thrombolysis. If there is no hemorrhage, CT angiogram looks for large vessel occlusion.

WHY NOT THE OTHERS

- (C) Aspirin is given within 24–48 hours, after hemorrhage is excluded.
- (D) For thrombolysis, BP needs only to be below 185/110.
- (E) Hemorrhage must be excluded before thrombolysis.

---

25 **(C) Right hemisphere**

Localizing a lesion · Step 1

Eyes deviated toward one side suggest a hemispheric lesion on that side, as in MCA stroke.

WHY NOT THE OTHERS

- (A) The eyes look toward a hemispheric lesion, here the right.
- (E) Dorsal midbrain lesions impair upgaze.

26 **D** **A swallow screen**

Acute ischemic stroke · Step 1

Supportive care comprises a swallow screen before oral intake, treating fever and glucose, and DVT prophylaxis, because dysphagia causes aspiration pneumonia.

WHY NOT THE OTHERS

- (B) A bubble echocardiogram looks for PFO.
- (E) Repeat imaging is for clinical worsening.

27 **A** **Delirium**

Ruling out mimics · Step 1

Sudden onset, fluctuating attention, and an acute illness indicate delirium, which needs a search for its cause.

WHY NOT THE OTHERS

- (B) Lewy body fluctuations occur over a chronic course with hallucinations and parkinsonism.
- (C) Depression causes low mood and poor effort.
- (D) Alzheimer disease develops over years.

28 **B** **Peripheral vertigo, with MRI if still unclear**

Ruling out mimics · Step 2

Isolated vertigo with a reassuring HINTS examination suggests peripheral vertigo, with MRI if unclear.

WHY NOT THE OTHERS

- (D) A reassuring HINTS examination points away from stroke.
- (E) Hemorrhage would usually cause other signs or show on CT.

29 **D** **Left V1**

Physiology · V and VII · Step 1

The corneal reflex runs V1 in and VII out. A left V1 lesion stops the afferent signal, so neither eye blinks; a CN VII lesion would stop the blink on that side only.

30 **B** **Once IOP is below 40 mmHg**

Glaucoma · Step 2

Acute angle closure is treated with acetazolamide, timolol, and apraclonidine, then IV mannitol if needed, and pilocarpine once IOP is below 40 mmHg; laser peripheral iridotomy follows.

WHY NOT THE OTHERS

- (C) At very high pressure the ischemic iris sphincter does not respond.
- (D) Pilocarpine constricts the pupil and helps open the angle.

31 **E** **Voltage-gated sodium channels are inactivated**

Physiology · Neurons and action potentials · Step 1

During the absolute refractory period, inactivated sodium channels prevent another spike; during the relative refractory period, a stronger stimulus is needed. These periods limit firing rate and keep conduction one-way.

32 **E** **DBS**

Typical Parkinson disease · Step 2

DBS treats refractory motor fluctuations.

WHY NOT THE OTHERS

- (A) Haloperidol blocks D2 receptors and worsens parkinsonism.
- (B) Tetrabenazine depletes dopamine.
- (C) Stopping levodopa would worsen parkinsonism.

33 **B** **CN IX afferent, CN X efferent**

Physiology · IX, X, XI, and XII · Step 1

CN IX carries touch from the back of the tongue and pharynx, forming the afferent limb, and CN X, from the nucleus ambiguus, supplies the pharynx, forming the efferent limb.

34 **E** **An early sign of ischemia**

Choosing imaging and EEG · Step 1

Early ischemia under 12 hours often looks normal or shows loss of gray-white differentiation, insular ribbon blurring, or lentiform nucleus effacement.

WHY NOT THE OTHERS

- (B) An old infarct is dark, like CSF.
- (D) SAH shows bright blood in the subarachnoid space.

35 **A** **CRVO**

Vision loss and red eye · Step 1

CRVO, mostly after age 50 with hypertension, glaucoma, diabetes, or hyperlipidemia, gives a blood and thunder fundus of dilated, tortuous veins and hemorrhages in all quadrants.

WHY NOT THE OTHERS

- (B) Detachment brings flashes, floaters, and a curtain.
- (D) CRAO shows a pale retina with a cherry-red spot.

36 **D** **Subarachnoid hemorrhage from a saccular aneurysm**

Subarachnoid hemorrhage · Step 1

PKD is one of the inherited conditions, with Ehlers-Danlos syndrome, fibromuscular dysplasia, coarctation of the aorta, and Marfan syndrome, linked with aneurysms, and these conditions need screening.

37 **C** **Rivastigmine**

Dementia with Lewy bodies · Step 2

Acetylcholinesterase inhibitors, rivastigmine, donepezil, and galantamine, offset cholinergic loss and improve cognition, function, and hallucinations in DLB.

38 **C** **Midbrain, upper limb flexion**

Physiology · Descending tracts · Step 1

The rubrospinal tract arises from the red nucleus, crosses in the midbrain, and favors upper limb flexion.

WHY NOT THE OTHERS

(E) Vestibulospinal and pontine reticulospinal tracts are uncrossed and excite extensors.

39 **D** **It slows Aβ clearance**

Alzheimer disease · Step 1

Decreased Aβ clearance is represented by APOE ε4, which slows clearance and raises the risk of late-onset disease; increased production comes from APP, PSEN1, and PSEN2 mutations and Down syndrome.

WHY NOT THE OTHERS

(A) The presenilins form the catalytic part of γ-secretase.

(B) Extra APP gene dosage occurs in Down syndrome.

40 **D** **The outermost layer of myelin and the oligodendrocyte surface**

MOGAD · Step 1

MOG is a transmembrane glycoprotein of the outermost myelin lamellae and oligodendrocyte surface, so antibodies reach it easily. MOGAD is a demyelinating disease, whereas AQP4-positive NMOSD is an astrocytopathy.

41 **E** **Vasospasm and delayed cerebral ischemia**

Stroke symptoms and complications · Step 2

Subarachnoid hemorrhage is complicated by rebleeding, highest early before the aneurysm is secured, and by vasospasm and delayed cerebral ischemia on days 3–14. New focal deficits on day 6 without new blood fit vasospasm.

WHY NOT THE OTHERS

(A) Hemorrhagic transformation follows ischemic stroke and would show blood within an infarct.

(D) Her aneurysm is secured, and CT shows no new blood.

42 **C** **Seed amplification assay**

Dementia with Lewy bodies · Step 1

CSF α-synuclein seed amplification assays have over 90% sensitivity and specificity, and only autopsy gives a definite diagnosis.

WHY NOT THE OTHERS

(A) Kappa free light chains support MS.

(B) Oligoclonal bands support MS.

(E) These support Alzheimer disease.

43 **A** **MOG-IgG**

MOGAD · Step 2

Bilateral optic neuritis with disc edema, an optic nerve lesion over half its length, and perineural optic sheath enhancement are supporting features of MOGAD, which affects men and women equally.

WHY NOT THE OTHERS

(D) NMOSD favors posterior and chiasmal lesions, mostly in women.

44 **C** **Idarucizumab**

Intracerebral hemorrhage · Step 2

Dabigatran is reversed with idarucizumab; apixaban or rivaroxaban with andexanet alfa or 4-factor PCC, heparin with protamine, and tPA with cryoprecipitate.

WHY NOT THE OTHERS

(A) Protamine reverses heparin.

(B) Vitamin K reverses warfarin.

(D) Cryoprecipitate reverses tPA.

(E) Andexanet reverses apixaban and rivaroxaban.

45 **C** **Central cord syndrome**

Stroke symptoms and complications · Step 1

Central cord syndrome, from hyperextension injury and syringomyelia, causes arm weakness

greater than leg, with the hands worst, cape-like pain and temperature loss, and sacral sparing.

WHY NOT THE OTHERS

- (B) Posterior cord syndrome causes sensory ataxia with preserved strength.
- (D) Anterior cord syndrome weakens all limbs below the lesion and spares vibration.
- (E) Hemisection gives one-sided weakness with opposite pain loss.

46 C Inhibition of inspiration to prevent overinflation

Physiology · Respiratory center · Step 1

Slowly adapting stretch receptors mediate the Hering-Breuer reflex, in which inflation inhibits inspiration. Irritant receptors cause cough, and juxtacapillary receptors respond to edema.

47 A Check and relieve bladder distension

Physiology · Cardiovascular center · Step 2

Autonomic dysreflexia follows SCI at or above T6. A stimulus below the lesion, usually a full bladder (85%), drives splanchnic vasoconstriction, while vagal bradycardia and flushing occur only above the lesion. Fixing the catheter or bladder is often life-saving.

48 E It dilated the pupil, letting the bunched iris narrow the angle

Glaucoma · Step 1

Sympathomimetics such as adrenaline, phenylephrine, and pseudoephedrine dilate the pupil, and the bunched iris base can close the angle in predisposed eyes, such as hyperopic eyes with a shallow anterior chamber.

49 D Prompt high-dose glucocorticoids

Localizing a lesion · Step 2

A new headache over 50, with jaw claudication, vision loss, or raised ESR and CRP, suggests GCA, which needs prompt high-dose glucocorticoids to prevent blindness.

WHY NOT THE OTHERS

- (A) Oxygen treats cluster headache.
- (B) This is the workup for a thunderclap headache.
- (C) Stopping analgesics treats medication overuse headache.

50 E IV methylprednisolone for 3–7 days

Multiple sclerosis · Step 2

An acute relapse is treated with IV methylprednisolone for 3–7 days, with plasma exchange if refractory.

WHY NOT THE OTHERS

- (A) Plasma exchange is for relapses refractory to steroids.
- (D) Baclofen treats spasticity.

Test B answers

Answer key | Explanations

1 A	11 B	21 E	31 C	41 E
2 B	12 B	22 E	32 B	42 A
3 C	13 A	23 E	33 D	43 B
4 D	14 C	24 A	34 C	44 A
5 B	15 B	25 A	35 E	45 C
6 E	16 C	26 E	36 D	46 B
7 C	17 C	27 D	37 B	47 D
8 C	18 D	28 E	38 E	48 D
9 D	19 A	29 C	39 D	49 A
10 B	20 D	30 A	40 E	50 A

1 A Ventilation-perfusion mismatch and the Haldane effect

Physiology | Respiratory center | Step 2

Oxygen raises CO_2 in COPD through ventilation-perfusion mismatch, as hypoxic vasoconstriction is reversed, and the Haldane effect, rather than loss of hypoxic drive, so saturations of 88–92% are targeted.

2 B It reaches cortex without a thalamic relay

Physiology | I, II, and VIII | Step 1

Olfactory receptor neurons, which regenerate for life, pass through the cribriform plate to mitral cells in the olfactory bulb, which project to piriform cortex, amygdala, and entorhinal cortex without a thalamic relay.

3 C Meningococcal vaccination

NMOSD | Step 2

Eculizumab blocks complement C5 and is given after meningococcal vaccination, since complement defends against Neisseria.

WHY NOT THE OTHERS

(A) HLA-B*1502 testing is for carbamazepine.

(E) JC virus testing is for natalizumab.

4 D Clarke nucleus, inferior cerebellar peduncle

Physiology | Ascending tracts | Step 1

The dorsal spinocerebellar tract arises in the Clarke nucleus (C8–L3) and ascends uncrossed through the inferior cerebellar peduncle; the ventral tract arises from spinal border cells and enters through the superior peduncle.

WHY NOT THE OTHERS

(B) This describes the ventral spinocerebellar tract.

5 B Relative preservation of the medial temporal lobe

Dementia with Lewy bodies | Step 1

In DLB, the medial temporal lobe is relatively preserved, so memory is spared early, while cortical involvement causes attention, executive, and visuospatial decline.

6 E GJB2

Hearing loss | Step 1

Nonsyndromic loss is autosomal recessive in 75–80%, so parents are usually unaffected carriers. DFNB1 mutations in GJB2 and GJB6, encoding connexins 26 and 30, are the most common genetic cause of severe to profound congenital deafness.

WHY NOT THE OTHERS

(B) PAX3 causes Waardenburg syndrome types 1 and 3, with pigment changes.

(D) Merlin mutations cause NF2, with bilateral vestibular schwannomas.

7 **C** **Pseudounipolar**

Physiology · Neurons and action potentials · Step 1

Most neurons, including motor neurons, are multipolar; some are bipolar; and sensory neurons of the dorsal root ganglia are pseudounipolar.

WHY NOT THE OTHERS

(A) Motor neurons are multipolar.
(E) Bipolar neurons have two separate processes.

8 **C** **MOG-IgG is found in only 0.4% of MS, so false positives result**

MOGAD · Step 1

Because MOG-IgG is found in only 0.4% of MS, testing everyone with possible MS yields false positives.

WHY NOT THE OTHERS

(A) A live cell-based assay uses serum.
(B) MS drugs appear ineffective in MOGAD.
(D) It is rare in MS.

9 **D** **GPi inhibition falls, causing chorea**

Physiology · Basal ganglia pathways · Step 1

In Huntington disease, indirect-pathway neurons die first, reducing GPi inhibition and causing chorea.

WHY NOT THE OTHERS

(A) This describes Parkinson disease.
(E) Dopamine loss is the mechanism in Parkinson disease.

10 **B** **Entacapone, a COMT inhibitor**

Typical Parkinson disease · Step 1

Entacapone is a COMT inhibitor that prolongs each levodopa dose; rasagiline is a selective MAO-B inhibitor, and pramipexole is a D3-selective dopamine agonist.

WHY NOT THE OTHERS

(A) Rasagiline inhibits MAO-B.
(C) Trihexyphenidyl is an anticholinergic.
(D) Amantadine is a weak NMDA antagonist.
(E) Pramipexole is a dopamine agonist.

11 **B** **A venous thrombus crossing a right-to-left shunt**

Acute ischemic stroke · Step 1

Paradoxical embolus occurs with ASD, PFO, and atrial septal aneurysm, where a right-to-left shunt lets a venous thrombus reach the arterial circulation. With no other cause and age under 60, PFO closure is considered.

12 **B** **PSP-P**

Atypical parkinsonism · Step 2

PSP-P may respond to levodopa and, like PSP-PAGF, allows survival of a decade or more, whereas PSP-RS leads to death about 7 years from onset.

WHY NOT THE OTHERS

(A) PSP-CBS is a cortical variant.
(E) PSP-RS is the classic form with early falls and gaze palsy.

13 **A** **Right jugular foramen**

Localizing a lesion · Step 2

CN IX, X, and XI together point to the jugular foramen. The right CN XI turns the head to the left and shrugs the right shoulder.

WHY NOT THE OTHERS

(C) The cerebellopontine angle involves CN V, VII, and VIII.
(E) The cavernous sinus holds CN III, IV, V1, V2, and VI.

14 **C** **Carotid endarterectomy**

Acute ischemic stroke · Step 2

Large vessel thrombus is treated with antiplatelet therapy and a high-intensity statin, with carotid endarterectomy for symptomatic ICA stenosis of 70–99%. Amaurosis fugax usually follows emboli from the carotid bifurcation.

15 **B** **Pneumonia**

Alzheimer disease · Step 1

Complications include aspiration pneumonia from dysphagia and infection, with pneumonia the most common cause of death.

WHY NOT THE OTHERS

(E) Amyloid angiopathy can cause lobar ICH, but it is not the most common cause of death.

16 **C** **4**

Intracerebral hemorrhage · Step 2

The ICH score gives 1 point for GCS 5–12 (2 for 3–4), 1 for age 80 or older, 1 for infratentorial origin, 1 for volume of 30 mL or more, and 1 for IVH. He

scores 1 + 1 + 0 + 1 + 1 = 4. The score estimates 30-day mortality.

17 **C** **No proven benefit, and behavior may worsen**

Frontotemporal dementia · Step 2

Cholinesterase inhibitors and memantine have no proven benefit in FTD, and donepezil can worsen behavior; SSRIs can improve behavior.

WHY NOT THE OTHERS

- (B) No treatment slows FTD.
- (D) Benefit has not been shown.

18 **D** **MRI for ARIA**

Alzheimer disease · Step 2

The anti-amyloid antibodies lecanemab and donanemab clear plaques in early disease, with monitoring for ARIA.

WHY NOT THE OTHERS

- (B) Bradycardia is an effect of acetylcholinesterase inhibitors.
- (C) JC virus status is checked before natalizumab.

19 **A** **Left thalamus**

Stroke symptoms and complications · Step 2

Central post-stroke pain often follows a thalamic infarct, though any sensory pathway lesion can cause it.

WHY NOT THE OTHERS

- (B) A right cerebellar lesion would cause ataxia on the right, not pain.
- (D) An occipital infarct causes visual field loss.
- (E) The caudate is not a sensory relay.

20 **D** **Lower renal sympathetic tone and ANP release, causing natriuresis**

Physiology · Cardiovascular center · Step 1

Cardiopulmonary receptors sense volume: stretch lowers renal sympathetic tone and releases ANP, causing natriuresis, while low volume raises sympathetic tone, RAAS activity, and vasopressin.

WHY NOT THE OTHERS

- (A) Stretch lowers renal sympathetic tone.
- (E) This follows low volume.

21 **E** **GABA-A receptor, chloride entry**

Physiology · Synapses and neurotransmitters · Step 1

Inhibitory postsynaptic potentials hyperpolarize the next neuron, as chloride enters through GABA-A receptors. Inputs sum at the axon hillock.

22 **E** **Cataract in the right eye**

Cataract · Step 1

Cataract causes monocular diplopia as light refracts through clear gaps in the lens.

WHY NOT THE OTHERS

- (A) Optic neuritis causes blurred vision with pain, not double images.
- (B) Nerve palsies cause binocular double vision.
- (D) Myasthenia causes binocular double vision.

23 **E** **Blood blocks CSF absorption at the arachnoid granulations**

Subarachnoid hemorrhage · Step 1

Blood blocks CSF absorption at the arachnoid granulations, causing hydrocephalus, which is treated with an EVD.

24 **A** **SAH can be ruled out without lumbar puncture**

Subarachnoid hemorrhage · Step 2

CT within 6 hours of onset has a sensitivity near 100%, so a normal scan can rule out SAH without a lumbar puncture; after 6 hours, a lumbar puncture follows a normal scan.

WHY NOT THE OTHERS

- (C) Angiography follows confirmed SAH.
- (E) A lumbar puncture follows a normal scan done after 6 hours.

25 **A** **Emotional and visceral responses to pain**

Physiology · Pain pathways and descending modulation · Step 1

The medial paleospinothalamic and archispinothalamic pathways project bilaterally to the reticular formation, PAG, and intralaminar thalamus, then to the cingulate, insula, and hypothalamus, carrying emotional and visceral responses.

WHY NOT THE OTHERS

- (C) Fine touch travels in the dorsal columns.
- (E) Location and intensity travel in the lateral tract.

26 **E** **Sorbitol accumulates and draws water into the lens**

Cataract · Step 1

Diabetes causes snowflake cortical opacities, because sorbitol accumulates and draws water into

the lens.

WHY NOT THE OTHERS

- (A) Urochrome collects in nuclear cataract.
- (C) This is phacolytic glaucoma.
- (D) Fibrous metaplasia occurs in subcapsular cataract.

27 (D) Prion disease

Ruling out mimics · Step 2

Rapid progression with myoclonus suggests prion disease, assessed with MRI and CSF.

28 (E) Corneal scraping, then urgent topical antibiotics

Vision loss and red eye · Step 2

Bacterial keratitis is a sight-threatening corneal infection, most often after contact lens wear, from organisms such as Staphylococcus aureus and Pseudomonas. Corneal scraping for Gram stain and culture precedes urgent topical antibiotics such as moxifloxacin.

WHY NOT THE OTHERS

- (C) Supportive care is for viral conjunctivitis, which spares vision.

29 (C) Immunoglobulin levels before treatment and yearly

NMOSD · Step 2

With RTX, immunoglobulin levels are checked before treatment and yearly, because low levels raise infection risk.

WHY NOT THE OTHERS

- (B) Titer does not predict severity or set the dose.

30 (A) Simple febrile seizure

Seizures · Step 2

Febrile seizures occur at 6 months to 5 years with fever of 38 °C or more and no CNS infection. Simple febrile seizures are a single generalized seizure of 15 minutes or less; complex ones have focal features, last 15 minutes or more, or recur within 24 hours.

31 (C) Medullary reticulospinal tract

Physiology · Descending tracts · Step 1

The medullary reticulospinal tract, mainly uncrossed, inhibits extensors, facilitates flexors, and contributes to autonomic and breathing control.

WHY NOT THE OTHERS

- (B) The pontine reticulospinal tract excites extensors.
- (E) The lateral vestibulospinal tract excites extensors.

32 (B) Increased perfusion at the focus

Choosing imaging and EEG · Step 2

Ictal SPECT shows increased perfusion at a seizure focus when injected within 90 seconds of onset, whereas FDG-PET shows low glucose use there between seizures.

WHY NOT THE OTHERS

- (C) Striatal uptake is measured by DAT SPECT.
- (D) Amyloid PET is for Alzheimer disease.
- (E) Low uptake at the focus is the interictal FDG-PET finding.

33 (D) Stop the suppressants and start vestibular rehabilitation

Vertigo · Step 2

Vestibular neuritis is treated with antiemetics, antihistamines such as meclizine, or benzodiazepines for about 3 days, then vestibular rehabilitation. Suppressants beyond a few days delay central compensation, and steroids lack evidence.

34 (C) Bilateral, high-frequency, downsloping SNHL

Hearing loss · Step 1

Presbycusis causes difficulty in noise and loss of high-pitched consonants, with a bilateral, high-frequency, downsloping curve.

WHY NOT THE OTHERS

- (A) A 4 kHz notch is noise-induced.
- (B) An air-bone gap marks conductive loss.
- (D) Low-frequency loss suggests Ménière disease.

35 (E) Propranolol

Motor signs and syndromes · Step 2

Postural tremor of both hands at 6–12 Hz, often familial, is essential tremor, treated with propranolol or primidone.

36 (D) Laser peripheral iridotomy

Glaucoma · Step 2

Acute angle closure is lowered with acetazolamide, timolol, and apraclonidine, then IV mannitol if needed and pilocarpine once IOP is below 40 mmHg. Laser peripheral iridotomy, which lets

aqueous humor bypass the pupil, is the treatment of choice.

WHY NOT THE OTHERS

- (C) Atropine dilates the pupil and can trigger or worsen angle closure.
- (E) Laser trabeculoplasty treats open-angle glaucoma.

37 (B) Vascular dementia

Ruling out mimics · Step 1

Stepwise decline with vascular risk factors suggests vascular dementia, shown on MRI.

WHY NOT THE OTHERS

- (A) Alzheimer disease declines gradually, with early memory loss.
- (D) Prion disease progresses over months with myoclonus.

38 (E) CN IX

Physiology · IX, X, XI, and XII · Step 1

CN IX carries pressure and oxygen signals from the carotid sinus and body that slow the heart, and CN X carries parasympathetic fibers to the heart.

WHY NOT THE OTHERS

- (B) CN X forms the efferent limb to the heart.

39 (D) Secondary progressive

Multiple sclerosis · Step 1

Gradual worsening after 10–15 years, with or without relapses, marks the secondary progressive course; primary progressive worsens from onset.

WHY NOT THE OTHERS

- (A) CIS is a first episode.
- (E) Primary progressive MS worsens from onset, without an earlier relapsing phase.

40 (E) Her respiratory muscles are tiring

Physiology · Respiratory center · Step 2

With intact drive, an asthma attack causes fast breathing and low CO_2, and a normal CO_2 signals tiring muscles.

WHY NOT THE OTHERS

- (C) A rising CO_2 during a continuing attack is a danger sign.
- (D) Drive is intact in asthma.

41 (E) D2 receptors on indirect-pathway neurons

Physiology · Basal ganglia pathways · Step 1

Dopamine-blocking drugs cause parkinsonism by blocking D2 receptors, which removes dopamine's inhibition of the indirect pathway.

WHY NOT THE OTHERS

- (C) Haloperidol is a D2 antagonist.

42 (A) Greater petrosal nerve to the lacrimal gland

Physiology · V and VII · Step 2

At the geniculate ganglion, the greater petrosal nerve leaves for the lacrimal gland; the chorda tympani carries taste and the stapedius branch prevents hyperacusis.

WHY NOT THE OTHERS

- (C) Only motor fibers leave the stylomastoid foramen.
- (D) Loss of the stapedius causes hyperacusis.
- (E) The chorda tympani carries taste and salivary fibers.

43 (B) Every fiber crosses before leaving the back of the midbrain

Physiology · III, IV, and VI · Step 1

Every CN IV fiber crosses before leaving the back of the midbrain, so a nuclear lesion weakens the opposite superior oblique.

WHY NOT THE OTHERS

- (A) CN VI runs free in the cavernous sinus.
- (D) CN III leaves through the interpeduncular fossa.
- (E) The CN VI nucleus is the horizontal gaze center.

44 (A) Observation, expecting it to resolve

Seizures · Step 2

Todd paralysis is temporary weakness after a focal or generalized tonic-clonic seizure, lasting minutes to 1–2 days and resolving without lasting damage. Mistaking it for a stroke can lead to unneeded thrombolysis.

WHY NOT THE OTHERS

- (E) A witnessed seizure with tongue biting points to Todd paralysis, a stroke mimic.

45 (C) C8–T1 injury from an apical lung tumor

Motor signs and syndromes · Step 2

Klumpke palsy, a C8–T1 injury from hyperabduction traction or an apical lung tumor, causes intrinsic hand weakness and clawing with medial forearm sensory loss, and Horner syndrome if sympathetic fibers are involved.

WHY NOT THE OTHERS

- (B) An ulnar lesion does not cause Horner syndrome or forearm numbness.
- (D) A medullary stroke does not cause hand wasting.
- (E) C5–C6 injury causes Erb palsy.

46 **B** **T2 shine-through, not acute infarct**

Choosing imaging and EEG · Step 2

Old edema or gliosis can be bright on DWI, T2 shine-through, but true acute infarct is dark on ADC.

WHY NOT THE OTHERS

(A) Acute infarct is dark on ADC.

(E) Enhancement is seen on T1 with gadolinium.

47 **D** **Glycine**

Physiology · Synapses and neurotransmitters · Step 1

GABA is the main inhibitory transmitter in the brain, and glycine is inhibitory mainly in the cord.

WHY NOT THE OTHERS

(A) Glutamate is the main excitatory transmitter.

48 **D** **Progressive supranuclear palsy**

Atypical parkinsonism · Step 2

The hallmark of PSP is vertical supranuclear gaze palsy, with downgaze affected first, and postural instability causes early backward falls with a stiff, upright gait.

49 **A** **Indirect-pathway medium spiny neurons projecting to the GPe**

Huntington disease · Step 1

Indirect-pathway neurons, which contain enkephalin and project to the GPe, are lost first, so the GPi inhibits the thalamus less and the released thalamus overdrives the cortex, causing chorea. Direct-pathway loss later causes rigidity and bradykinesia.

50 **A** **Alzheimer disease**

Frontotemporal dementia · Step 2

Word-finding pauses with poor repetition favor logopenic PPA, which is usually caused by Alzheimer disease.

WHY NOT THE OTHERS

(C) DLB brings hallucinations, parkinsonism, and fluctuations.

Test C answers

Answer key | Explanations

1 A	11 A	21 E	31 D	41 C
2 A	12 C	22 C	32 D	42 B
3 B	13 A	23 B	33 C	43 E
4 A	14 E	24 C	34 E	44 B
5 B	15 A	25 A	35 D	45 B
6 D	16 E	26 B	36 C	46 A
7 B	17 B	27 A	37 E	47 E
8 E	18 B	28 D	38 E	48 D
9 E	19 D	29 A	39 C	49 C
10 D	20 C	30 C	40 D	50 D

1 A Westphal variant of juvenile Huntington disease

Huntington disease | Step 1

Juvenile disease, before age 20, causes rigidity and bradykinesia (the Westphal variant), dystonia, learning difficulty, and seizures in 30–50% before age 10, usually with over 55–60 repeats.

2 A JC virus status

Multiple sclerosis | Step 2

Natalizumab can cause PML, so JC virus status is checked first.

WHY NOT THE OTHERS

(E) Bradycardia on starting is a concern with fingolimod, through S1P receptors on the SA and AV nodes.

3 B The GPi and SNr tonically inhibit the VA and VL thalamus

Physiology | Basal ganglia pathways | Step 1

The GPi and SNr form the output and tonically inhibit the VA and VL thalamus, which relays to the motor cortex.

WHY NOT THE OTHERS

(E) The STN excites the GPi.

4 A Right subthalamic nucleus

Motor signs and syndromes | Step 1

Hemichorea and hemiballismus classically follow a lesion of the contralateral subthalamic nucleus, most often from ischemic stroke over 65, and tend to resolve.

5 B Pyridostigmine

Physiology | Synapses and neurotransmitters | Step 2

Myasthenia gravis targets postsynaptic nicotinic AChR or MuSK, causing fatigable weakness with ptosis and diplopia in 85%. Pyridostigmine, a cholinesterase inhibitor, comes first, then immunosuppression.

6 D Lens proteins leak and macrophages clog the trabecular meshwork

Cataract | Step 2

In phacolytic glaucoma, a hypermature lens leaks proteins, and macrophages clog the trabecular meshwork while the angle stays open; it is treated by removing the lens.

WHY NOT THE OTHERS

(A) This is phacomorphic glaucoma, with a closed angle.

(B) Pupillary block closes the angle.

7 **B** **Left CN XII**

Physiology · IX, X, XI, and XII · Step 1

A CN XII lesion deviates the tongue toward the lesion, with wasting and fasciculations.

8 **E** **Foramen ovale**

Physiology · V and VII · Step 1

V1 passes through the superior orbital fissure, V2 through the foramen rotundum, and V3, which supplies the lower jaw and muscles of mastication, through the foramen ovale.

9 **E** **Increases uveoscleral outflow**

Glaucoma · Step 1

Prostaglandin analogs such as latanoprost increase uveoscleral outflow and are first line; they lengthen eyelashes and darken the iris.

WHY NOT THE OTHERS

(B) This is how carbonic anhydrase inhibitors work.
(C) This is how pilocarpine works.
(D) This is how timolol works.

10 **D** **NMOSD**

NMOSD · Step 1

CSF OCB are usually absent in NMOSD, and their presence is a red flag. Over 50 white cells/μL, or neutrophils or eosinophils, favor NMOSD over MS.

11 **A** **Neurosyphilis**

Physiology · I, II, and VIII · Step 1

Argyll Robertson pupils constrict to a near target but not to light and mark neurosyphilis.

12 **C** **Left-sided vibration and position loss**

Physiology · Ascending tracts · Step 1

Medial medullary syndrome damages the medial lemniscus, which has already crossed, causing opposite-side vibration and position loss.

WHY NOT THE OTHERS

(B) Cape-like loss follows syringomyelia.
(E) The medial lemniscus carries fibers from the opposite side.

13 **A** **Delirium**

Alzheimer disease · Step 2

Delirium is favored by sudden onset, fluctuating attention, and an acute illness, and prompts a search for the cause.

14 **E** **Fasciculus gracilis**

Physiology · Ascending tracts · Step 1

Large DRG fibers ascend on the same side, gracilis from below T6 and cuneatus from T6 up.

WHY NOT THE OTHERS

(A) The spinothalamic tract carries pain and temperature.
(B) Cuneatus carries fibers from T6 up.

15 **A** **Petrosal and nodose ganglia**

Physiology · Cardiovascular center · Step 1

Carotid sinus afferents travel in CN IX with cell bodies in the petrosal ganglion, and aortic afferents in CN X with cell bodies in the nodose ganglion.

WHY NOT THE OTHERS

(B) The order is reversed.
(C) These are sympathetic ganglia.

16 **E** **A seizure spreading along the motor cortex**

Localizing a lesion · Step 1

Symptoms spreading to the next body part are a Jacksonian march, a seizure spreading along the cortex.

WHY NOT THE OTHERS

(A) A visual aura starts in the vision.
(B) A cord lesion spares the face.
(C) Ischemia causes sudden loss of function rather than a spreading positive symptom.

17 **B** **Allow it up to 220/120**

Acute ischemic stroke · Step 2

Without thrombolysis, BP is allowed up to 220/120 in ischemic stroke, whereas it is kept below 185/110 for thrombolysis and actively lowered in intracerebral hemorrhage.

WHY NOT THE OTHERS

(C) A systolic target of 140 is for intracerebral hemorrhage.
(E) Nimodipine prevents delayed ischemia after subarachnoid hemorrhage.

18 **B** **CN IV**

Localizing a lesion · Step 1

A CN IV palsy, with the head tilted away, often follows trauma.

WHY NOT THE OTHERS

(A) CN VI palsy causes horizontal double vision.
(C) CN III palsy turns the eye down and out with ptosis.
(D) An MLF lesion causes INO.

19 (D) Astrocyte

NMOSD · Step 1

AQP4-IgG binds AQP4 on astrocyte end-feet and kills astrocytes through the classical complement pathway, so demyelination is secondary, with oligodendrocytes and axons damaged as bystanders.

WHY NOT THE OTHERS

(B) Oligodendrocytes are the target in MOGAD and are damaged only secondarily in NMOSD.

20 (C) Video EEG

Seizures · Step 1

Closed eyes, side-to-side head movement, asynchronous limb movement, and no post-ictal state suggest psychogenic non-epileptic seizures, confirmed by video EEG.

WHY NOT THE OTHERS

(D) Prolactin also rises after convulsive syncope, and a normal level does not confirm PNES.

21 (E) Topical antibiotic drops such as chloramphenicol

Vision loss and red eye · Step 1

Bacterial conjunctivitis causes mucopurulent discharge with lids stuck together, often bilaterally, with normal vision; it is often self-limited, and topical antibiotics shorten it.

WHY NOT THE OTHERS

(B) Antihistamines treat allergic conjunctivitis.
(C) Steroids treat anterior uveitis.

22 (C) CRPS

Physiology · Pain pathways and descending modulation · Step 2

CRPS follows minor injury or surgery with severe limb pain, allodynia, and changes in sweating and blood flow, as inflammation sensitizes nociceptive neurons and sympathetic drive sustains the pain.

WHY NOT THE OTHERS

(D) Carpal tunnel syndrome causes tingling in the median nerve distribution without sweating changes.

23 (B) Phenytoin

Seizures · Step 1

Phenytoin causes gingival hyperplasia, nystagmus and ataxia at high levels, SJS, and fetal hydantoin syndrome, and follows zero-order kinetics at high doses.

WHY NOT THE OTHERS

(D) Valproate causes hepatotoxicity and neural tube defects.
(E) Carbamazepine causes agranulocytosis and hyponatremia.

24 (C) ECG and orthostatic BP

Ruling out mimics · Step 2

A trigger, lightheadedness, brief jerks, and rapid recovery suggest syncope, assessed with ECG and orthostatic BP.

25 (A) Low-dose quetiapine

Dementia with Lewy bodies · Step 2

Antipsychotics are avoided whenever possible, because serious reactions occur in up to 50%; if one is essential, low-dose quetiapine is used, and haloperidol, risperidone, olanzapine, and aripiprazole are generally avoided.

WHY NOT THE OTHERS

(B) Haloperidol is generally avoided.
(C) Aripiprazole is generally avoided.
(D) Olanzapine is generally avoided.
(E) Risperidone is generally avoided.

26 (B) The upper face receives corticobulbar input from both hemispheres

Physiology · Descending tracts · Step 1

Corticobulbar input is mostly bilateral, and the upper face receives input from both hemispheres, so a cortical lesion spares the forehead.

27 (A) Systolic below 160 with IV nicardipine or labetalol

Subarachnoid hemorrhage · Step 2

BP is controlled with IV nicardipine or labetalol until the aneurysm is secured, keeping systolic below 160.

WHY NOT THE OTHERS

(B) That threshold is for ischemic stroke without reperfusion.
(C) That threshold applies before thrombolysis.

(E) This is the long-term target after hypertensive ICH.

28 **(D) Right anterior temporal lobe**

Frontotemporal dementia · Step 1

Right, nondominant, anterior temporal atrophy causes prosopagnosia, getting lost, behavior change, and memory loss.

WHY NOT THE OTHERS

(A) Left inferior frontal atrophy causes the nonfluent variant.

(C) Left anterior temporal atrophy causes the semantic variant.

29 **(A) Yes, the conus lesion is a supporting feature**

MOGAD · Step 2

A low positive titer of 1:10 to below 1:100 requires negative AQP4-IgG and at least one supporting feature, such as a conus lesion, with better diagnoses excluded.

WHY NOT THE OTHERS

(B) Myelitis is one of the most common syndromes.

(C) Low titers are accepted with a supporting feature.

(D) Better diagnoses, including MS, must always be excluded.

30 **(C) Retained bicarbonate blunts the central chemoreceptors**

Physiology · Respiratory center · Step 1

In obesity hypoventilation, chest wall mechanics cause hypoventilation, and the kidneys retain bicarbonate, which blunts the sensitivity of the central chemoreceptors, so hypercapnia persists.

31 **(D) Blocks voltage-gated sodium channels in the nerve**

Physiology · Neurons and action potentials · Step 1

Local anesthetics such as lidocaine block voltage-gated sodium channels in nerves, so pain signals fail to conduct. Phenytoin, carbamazepine, and lamotrigine also reduce sodium influx.

32 **(D) Ventral pontine (Millard-Gubler) syndrome**

Stroke symptoms and complications · Step 1

Ventral pontine (Millard-Gubler) syndrome causes contralateral hemiplegia, ipsilateral CN VI and CN VII palsy with the forehead included, and horizontal diplopia.

WHY NOT THE OTHERS

(A) AICA syndrome spares strength and adds hearing loss.

(B) Foville syndrome causes a horizontal gaze palsy rather than an isolated abduction deficit.

(C) Wallenberg syndrome causes no hemiparesis.

(E) Weber syndrome causes a CN III palsy.

33 **(C) Antioxidant and zinc supplements**

Vision loss and red eye · Step 1

Dry AMD, most AMD, causes gradual central loss with drusen; AREDS antioxidant and zinc supplements slow it, smoking is stopped, and an Amsler grid detects conversion to wet AMD.

WHY NOT THE OTHERS

(A) Retinopexy repairs a retinal detachment.

(B) Anti-VEGF treats wet AMD with fluid or new vessels.

34 **(E) Orthostatic hypotension**

Physiology · Cardiovascular center · Step 2

Orthostatic hypotension is a sustained fall of 20 mmHg systolic or 10 mmHg diastolic within 3 minutes of standing, as blood pools in the legs and splanchnic veins, and occurs in Parkinson disease, MSA, and DLB.

WHY NOT THE OTHERS

(A) In POTS, HR rises by 30 or more without orthostatic hypotension.

35 **(D) Huntington disease**

Choosing imaging and EEG · Step 1

Huntington disease shows caudate atrophy, even before symptoms.

WHY NOT THE OTHERS

(A) PD has a normal structural MRI, with reduced DAT uptake.

(C) PSP shows midbrain atrophy, the hummingbird sign.

(E) MSA shows the hot cross bun sign.

36 **(C) Labyrinthitis**

Vertigo · Step 1

Vestibular neuritis spares hearing, whereas labyrinthitis adds unilateral hearing loss and tinnitus.

WHY NOT THE OTHERS

(A) BPPV attacks last under a minute.

(B) Vestibular neuritis spares hearing.

(D) Ménière attacks last 20 minutes to 12 hours.

37 **E** **Less salt, caffeine, and alcohol, with a thiazide or betahistine**

Vertigo · Step 2

Ménière disease, from endolymphatic hydrops, causes attacks of 20 minutes to 12 hours with fluctuating low-frequency loss, tinnitus, and fullness. It is treated with less salt, caffeine, and alcohol, thiazides or betahistine, intratympanic steroids, and intratympanic gentamicin if refractory.

38 **E** **Hemorrhagic metastasis**

Intracerebral hemorrhage · Step 2

Malignancy, including melanoma, RCC, choriocarcinoma, and thyroid carcinoma, disrupts the BBB and bleeds, shown as an enhancing mass on MRI with contrast.

WHY NOT THE OTHERS

- (B) Mycotic aneurysms follow endocarditis.
- (C) Hypertensive bleeds are deep.
- (D) Amyloid angiopathy shows lobar microbleeds without a mass.

39 **C** **Hearing becomes bilateral above the cochlear nuclei**

Physiology · I, II, and VIII · Step 1

Hearing becomes bilateral at the trapezoid body, so one-sided deafness localizes to the ear, nerve, or cochlear nucleus.

40 **D** **Common peroneal nerve at the fibular head**

Motor signs and syndromes · Step 1

Common peroneal compression at the fibular head causes foot drop with weak dorsiflexion and eversion and a numb dorsum. Inversion, supplied by the tibial nerve, is spared, which separates it from L5 radiculopathy, the most common cause of foot drop.

WHY NOT THE OTHERS

- (E) An L5 lesion also weakens inversion.

41 **C** **Add entacapone**

Typical Parkinson disease · Step 2

As neurons are lost, 'on' time shortens, and entacapone, a COMT inhibitor that prolongs each levodopa dose, or extended-release levodopa is added. DBS treats refractory motor fluctuations.

42 **B** **CSF oligoclonal bands**

Multiple sclerosis · Step 2

The 2024 criteria need dissemination in space, with plaques in at least 2 of the 5 sites, here met with the optic nerve, periventricular, and juxtacortical sites. Dissemination in time is missing, but CSF oligoclonal bands, kappa free light chains, 4 or more sites, or 6 plaques with a central vein sign can substitute for time.

WHY NOT THE OTHERS

- (A) Positive AQP4 antibody points to NMOSD rather than MS.

43 **E** **Ventilation tubes**

Hearing loss · Step 2

Otitis media with effusion is the most common cause of acquired hearing loss in children, and ventilation tubes treat persistent effusion.

WHY NOT THE OTHERS

- (A) This is an acquired conductive loss.
- (B) Stapes surgery treats otosclerosis.
- (C) Prednisone is for sudden SNHL.
- (D) Implants are for severe bilateral SNHL.

44 **B** **It blocks peripheral conversion of levodopa to dopamine**

Typical Parkinson disease · Step 1

Levodopa crosses the BBB and becomes dopamine, and carbidopa blocks its peripheral conversion; together they are the most effective treatment.

WHY NOT THE OTHERS

- (D) Rasagiline is the MAO-B inhibitor.
- (E) Entacapone is the COMT inhibitor.

45 **B** **Ischemia affects the core of the nerve, sparing the surface pupil fibers**

Physiology · III, IV, and VI · Step 1

Pupil fibers ride on the surface of CN III, so a PComm aneurysm compressing the nerve dilates the pupil early, whereas a diabetic infarct spares it.

46 **A** **Left posterior cerebral artery**

Stroke symptoms and complications · Step 1

PCA syndrome usually causes no weakness, with contralateral homonymous hemianopia with macular sparing. Alexia without agraphia occurs with dominant (left) lesions, and prosopagnosia with non-dominant lesions.

WHY NOT THE OTHERS

(C) MCA syndrome causes face and arm weakness.

(E) A right PCA stroke causes a left hemianopia and prosopagnosia.

47 E Charcot-Bouchard aneurysms in penetrating arterioles

Intracerebral hemorrhage · Step 1

Chronic hypertension causes lipohyalinosis and Charcot-Bouchard aneurysms in penetrating arterioles, which bleed in the basal ganglia, thalamus, pons, and cerebellum.

WHY NOT THE OTHERS

(A) Saccular aneurysms cause subarachnoid hemorrhage.

(D) Amyloid angiopathy causes lobar bleeds.

48 D FLAIR

Choosing imaging and EEG · Step 1

FLAIR makes lesions bright with fluid made dark, showing periventricular plaques, SAH in the sulci, and infarcts after 4.5–6 hours.

WHY NOT THE OTHERS

(A) T1 makes fat and gadolinium bright.

(C) MRA shows flowing blood.

(E) SWI shows blood products as dark.

49 C Multiple system atrophy

Atypical parkinsonism · Step 1

MSA is an α-synucleinopathy with GCIs in oligodendrocytes. In PD and DLB, α-synuclein collects in neurons, and PSP and CBD are 4R tauopathies.

50 D Mood and suicidal thoughts

Huntington disease · Step 2

Tetrabenazine inhibits VMAT2, reducing dopamine release and lessening chorea. It can cause depression, suicidality, fatigue, and akathisia, so mood and suicidal thoughts are monitored.